Some combinatorial processes

An illustrated mathematical notebook

Jacob Richey

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About this version. The original version of these notes was written by Jacob Richey. ChatGPT Astra revised the calculations and developed some of the ideas further.

A pile of unmatched socks grows and then disappears. A row of seats fills until every vacancy is blocked. A record survives until a larger observation replaces it. Each construction turns a sequence of small decisions into a random object with a shape, a time scale, and a useful statistic.

The sixteen sections below follow these objects from exact counting to limiting processes. Independent clocks expose some dependencies; stopping times and generating functions organize others. Pictures show sample paths, configurations, and the distributions suggested by the calculations. The questions at the ends of the arguments identify further calculations and extensions to explore. Classical results are stated with attribution; proofs developed here explain or extend particular calculations without a claim of priority. Questions are proposed directions, with their literature status specified where it is known.

Contents

1 The sock process **2**
2 The dish process **6**
3 The coupon collector process **8**
4 Non-adjacent uniform placements **11**
5 Adjacent occurrences in a random permutation **16**
6 Hitting times of sums **18**
7 Heatseekers **21**
8 The continuous coupon collector process **24**
9 Asymptotics of hitting times and expected value **26**
10 Number of maxima of iid random variables **28**
11 Functional limit for the extreme-value process **31**
12 Ruler distribution **34**
13 Random decreasing sequences **36**
14 Fisherman's dilemma **38**
15 Limits of multiplicative functions on $\mathbb{N}$ **40**
16 Egyptian fractions: random inputs and the case $4/n$ **43**

1 The sock process

Draw socks one at a time from a basket containing n pairs. Keep each unmatched sock on the table and remove a pair as soon as its second member appears. The table starts empty and ends empty; halfway through, it holds about $n/2$ socks. The first question is the shape of this rise and fall. The second is how the fluctuations are constrained by having drawn a fixed number of socks.

The model has a substantial history. Steinsaltz's random-time-change theorem [1, Proposition 4.1] covers the functional Gaussian limit, including much more general group sizes. The fixed-size case below is a specialization, with a direct clock proof that makes the covariance visible. His earlier work [2] also identifies the finer fluctuations near the maximum. These are classical inputs to the story.

Write X_k for the number on the table after draw k. Sampling is uniform without replacement, and $X_0 = X_{2n} = 0$.

More generally, fix an integer $l \geq 2$ and draw from n groups of l objects. Remove a group as soon as all its members have been drawn. If Y_k counts completed groups, then $X_k = k - lY_k$. Write $(a)_r = a(a-1)\cdots(a-r+1)$, with $(a)_0 = 1$. An indicator calculation gives

$$\mathbb{E}Y_k = n\frac{(k)_l}{(ln)_l}, \tag{1.1}$$

$$\mathbb{E}X_k = k - ln\frac{(k)_l}{(ln)_l}, \tag{1.2}$$

$$\mathrm{Var}(X_k) = l^2\left\{ n\frac{(k)_l}{(ln)_l} + n(n-1)\frac{(k)_{2l}}{(ln)_{2l}} - n^2\left(\frac{(k)_l}{(ln)_l}\right)^2 \right\}. \tag{1.3}$$

The variance formula is written for $n \geq 2$; for $n = 1$ the process is deterministic. For fixed $0 < t < 1$ and $k = \lfloor lnt \rfloor$,

$$\frac{\mathbb{E}X_k}{ln} \longrightarrow t - t^l, \qquad \frac{\mathrm{Var}(X_k)}{l^2 n} \longrightarrow v_l(t) := t^l(1-t^l) - lt^{2l-1}(1-t). \tag{1.4}$$

The limiting mean $t - t^l$ reaches its maximum at $t_l = l^{-1/(l-1)}$, with height $(1 - 1/l)t_l$. Larger groups move the peak toward the end of the sample: $t_l = 1 - (\log l)/(l-1) + o((\log l)/l)$. The fluctuation scale records a second effect. Fixing the total number drawn creates negative dependence between groups, represented by the second term in v_l. It remains after the order-n^2 terms in the variance cancel.

There is also an exact covariance formula. If $0 \leq k \leq m \leq ln$ and $m \geq l$, then

$$\mathbb{E}(Y_k Y_m) = n\frac{(k)_l}{(ln)_l} + n(n-1)\frac{(k)_l}{(ln)_l}\frac{(m-l)_l}{(ln-l)_l}. \tag{1.5}$$

Indeed, conditional on one group being complete by draw k, all its l members lie among the first m draws. The remaining $m - l$ positions form a uniform sample from the other $ln - l$ objects. If $m < l$, both counts vanish. Subtracting the product of the means in (1.5) gives the covariance of Y_k, Y_m, and multiplication by l^2 gives that of X_k, X_m.

Proposition 1.1 (Steinsaltz: fixed-size specialization)**.** *For fixed l, linearly interpolate the process*

$$Z_n(k/(ln)) = \frac{X_k - \mathbb{E}X_k}{l\sqrt{n}}.$$

Then Z_n converges in distribution in $C[0,1]$ to a centered continuous Gaussian process Z with, for $0 \leq s \leq t \leq 1$,

$$\mathrm{Cov}(Z(s), Z(t)) = s^l(1-t^l) - ls^l t^{l-1}(1-t). \tag{1.6}$$

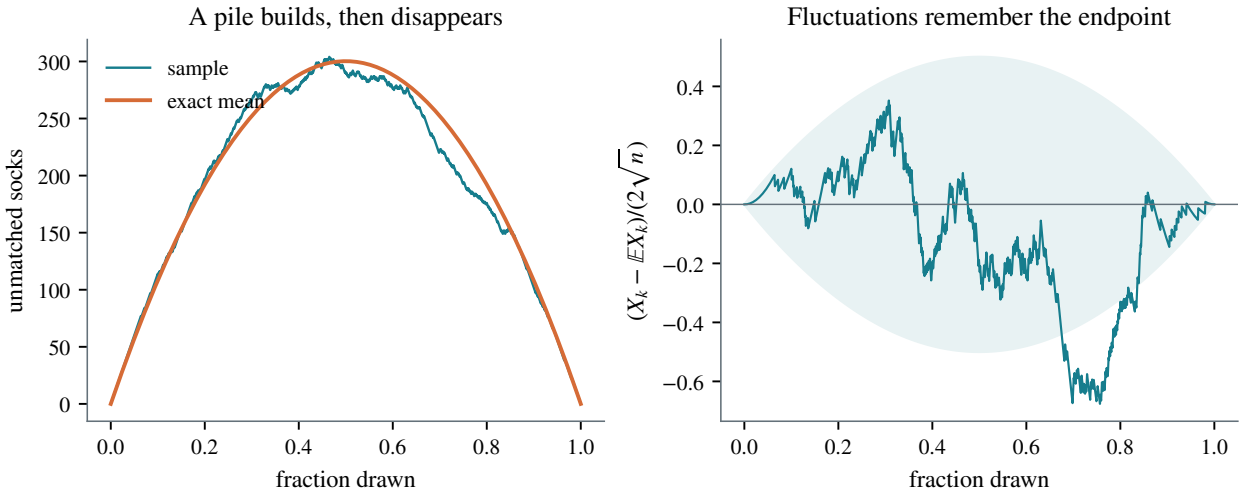


Figure 1: A uniform ordering of 600 pairs. Left: the number of unmatched socks and its exact mean. Right: the same sample after centering and scaling. The shaded region is twice the limiting pointwise standard deviation $t(1-t)$ on either side of zero; it indicates scale rather than simultaneous coverage.

Proof. Give the members of group i independent uniform clocks $U_{i1}, \dots, U_{il}$. Ordering all ln clocks produces a uniform ordering of the objects. Put

$$Q_i(t) = 1\{\max_j U_{ij} \le t\}, \quad C_i(t) = \sum_{j=1}^{l} 1\{U_{ij} \le t\},$$

and

$$A_n(t) = n^{-1/2} \sum_i (Q_i(t) - t^l), \qquad B_n(t) = n^{-1/2} \sum_i (C_i(t) - lt).$$

These are finite collections of empirical distribution functions of iid group vectors. The empirical-process central limit theorem gives joint convergence to continuous centered Gaussian processes (A, B): the indexing sets are nested intervals for each coordinate and nested lower cubes for the group maximum. Equivalently, each coordinate is an ordinary empirical bridge; coordinatewise tightness and the multivariate central limit theorem give joint convergence.

Let $T_n(t)$ be the $\lfloor lnt \rfloor$th clock, with $T_n(0) = 0$. The empirical distribution of all clocks is $F_n(u) = u + B_n(u)/(l\sqrt{n})$, and $|F_n(T_n(t)) - t| \le 1/(ln)$. Its uniform convergence to the identity and the asymptotic equicontinuity of B_n imply, uniformly in t,

$$\sqrt{n}(T_n(t) - t) = -B_n(t)/l + o_{\mathbb{P}}(1).$$

Consequently

$$\frac{Y_{\lfloor lnt \rfloor} - nt^l}{\sqrt{n}} = A_n(t) - t^{l-1} B_n(t) + o_{\mathbb{P}}(1).$$

The exact expectation differs from nt^l by a uniformly bounded quantity. The interpolation changes Z_n by at most $O(n^{-1/2})$. Thus the limit is $Z(t) = -A(t) + t^{l-1}B(t)$. For $s \le t$, the covariances of a single group are

$$\begin{aligned} \operatorname{Cov}(Q(s), Q(t)) &= s^l(1 - t^l), & \operatorname{Cov}(C(s), C(t)) &= ls(1-t), \\ \operatorname{Cov}(Q(s), C(t)) &= ls^l(1-t), & \operatorname{Cov}(C(s), Q(t)) &= lst^{l-1}(1-t). \end{aligned}$$

Substitution gives (1.6). □

For $l = 2$, the covariance simplifies to $s^2(1-t)^2$ for $s \le t$. In particular, the variance is $t^2(1-t)^2$. One representation in terms of standard Brownian motion is

$$Z(t) \stackrel{d}{=} (1-t)^2 B\left(\frac{t^2}{(1-t)^2}\right), \quad 0 \le t < 1, \qquad Z(1) = 0.$$

This expresses the limiting fluctuations as a time and amplitude change of Brownian motion.

For pairs, one can also count the one-time distribution directly:

$$\mathbb{P}(X_k = m) = \frac{2^m \binom{n}{(k-m)/2} \binom{n-(k-m)/2}{m}}{\binom{2n}{k}}, \tag{1.7}$$

when the binomial coefficients and parity are admissible, and zero otherwise. The m unmatched pairs contribute a choice of one of their two socks. For asymptotic calculations, the relevant Stirling formula is

$$\binom{N}{cN} \sim \frac{[c^{-c}(1-c)^{-(1-c)}]^N}{\sqrt{2\pi N c(1-c)}},$$

along integers with c bounded away from 0 and 1.

The pair process is Markov: its next step is -1 with probability $X_k/(2n-k)$ and $+1$ otherwise. If $D_k = X_k - \mathbb{E}X_k$, then

$$\frac{D_k}{(2n-k)(2n-k-1)}, \qquad 0 \le k \le 2n-2,$$

is a martingale. This follows by dividing $\mathbb{E}(D_{k+1} \mid \mathcal{F}_k) = (1-2/(2n-k))D_k$ by the next denominator. For $l > 2$, the numbers of partially completed groups of each size provide a Markov state.

1.1 Paths and the maximum

An exact path formula complements the Gaussian limit. A Dyck path here is a nonnegative nearest-neighbor path $x_0 = 0, x_1, \dots, x_{2n} = 0$. Korbel and Mörters [3] give its probability:

Theorem 1.2 (Korbel–Mörters, in path notation)**.** *For n pairs and any such path,*

$$\mathbb{P}((X_k)_{k=0}^{2n} = (x_k)_{k=0}^{2n}) = \frac{1}{(2n-1)!!} \prod_{k : x_k = x_{k-1}-1} x_{k-1}. \tag{1.8}$$

Proof. The pair endpoints induce a uniform perfect matching of the $2n$ draw positions. At a downward step there are x_{k-1} unmatched opening positions to close. Multiplying these choices counts the matchings with the given path, out of $(2n-1)!!$ matchings in total. □

Thus $UDUD$ has probability $1/3$ for two pairs, while $UUDD$ has probability $2/3$. Counting paths without these weights changes the model.

The weights also compute the maximum exactly. Let T_h be the matrix on $\{0, \dots, h\}$ with $(T_h)_{j,j+1} = 1$ and $(T_h)_{j,j-1} = j$, and all other entries zero. If $M_n = \max_k X_k$, matrix multiplication sums the weights of paths confined to this interval, so

$$\mathbb{P}(M_n \le h) = \frac{(T_h^{2n})_{0,0}}{(2n-1)!!}. \tag{1.9}$$

The functional limit already determines the first fluctuation scale:

Proposition 1.3. *For pairs,* $(M_n - n/2)/\sqrt{n} \longrightarrow_d N(0, 1/4)$.

Proof. The deterministic mean divided by n converges uniformly to $2t(1-t)$, whose unique maximum is at $1/2$. Proposition 1.1 gives a tight, asymptotically equicontinuous centered process, so any maximizing time t_n tends in probability to $1/2$. The exact mean is maximized at draw n. Consequently

$$0 \le M_n - X_n \le (X_{2nt_n} - \mathbb{E}X_{2nt_n}) - (X_n - \mathbb{E}X_n) = o_{\mathbb{P}}(\sqrt{n}).$$

At time $1/2$, the limiting variance for $X_n/\sqrt{n}$ is $1/4$. Also $\mathbb{E}X_n = n^2/(2n-1) = n/2 + O(1)$. □

Steinsaltz's maximum theorem goes further: the excess above the midpoint value has order $n^{1/3}$, and the maximizing draw fluctuates on the $n^{2/3}$ scale [1, Section 4.1]. The local limit is Brownian motion with a downward quadratic drift. The exponents can be read off before the theorem: moving h draws from the middle loses mean height of order h^2/n, while local random-walk fluctuations have size $\sqrt{h}$. Their balance is $h \asymp n^{2/3}$, with height scale $n^{1/3}$.

Question 1.4. When the common group size grows with n, where do the fixed-size Gaussian and maximum approximations cease to be uniform? Variable group sizes with bounded moments are already treated by Steinsaltz; a growing largest group requires new control of that approximation.

2 The dish process

A sink accumulates dishes one at a time and is cleaned in occasional batches. The cleaning rule determines whether the pile has a stationary size, makes ever longer excursions, or escapes. A drift calculation suggests the regimes; return times distinguish them.

Let $X_n \geq 0$ be the amount in the sink. Choose $p : [0, \infty) \to [0, 1]$ and $0 \leq \beta(x) \leq x$, and set

$$X_{n+1} = \begin{cases} X_n + 1, & \text{with probability } 1 - p(X_n), \\ X_n - \beta(X_n), & \text{with probability } p(X_n). \end{cases} \tag{2.1}$$

Here $\beta(x)$ is the amount cleaned, so the amount left is $x - \beta(x)$.

The conditional drift is

$$\mathbb{E}(X_{n+1} - X_n \mid X_n = x) = 1 - p(x)(1 + \beta(x)). \tag{2.2}$$

Thus the zero-drift equation is $p(x)(1 + \beta(x)) = 1$. It identifies a balance of expected increments; recurrence and convergence require more information.

Example 2.1. Take $p(x) = 1/2$ and $\beta(x) = x$ on $\mathbb{Z}_{\geq 0}$. At each step we either add one dish or empty the sink. Its stationary law is $\pi(j) = 2^{-j-1}$, since $\pi(0) = \frac{1}{2} \sum_j \pi(j)$ and $\pi(j) = \frac{1}{2}\pi(j-1)$ for $j \geq 1$. Independent reset coins imply infinitely many returns to zero, while arbitrarily long runs of additions give $\limsup X_n = \infty$ almost surely. The stationary law describes the proportion of time spent at each height; individual paths keep making fresh excursions.

Cleaning one dish at a time gives an explicit phase diagram. Set $p(0) = 0$ and take $\beta(x) = 1$, $p(x) = p \in (0, 1)$ at positive integers. The chain moves from 0 to 1 and, above zero, moves down with probability p and up with probability $1 - p$. For $p > 1/2$, detailed balance gives

$$\pi(0) = \frac{2p - 1}{2p}, \qquad \pi(j) = \frac{\pi(0)}{p} \left(\frac{1 - p}{p} \right)^{j-1}, \quad j \geq 1.$$

For $p = 1/2$ the chain returns to zero almost surely but has no summable stationary weights. For $p < 1/2$ it tends to infinity almost surely. Indeed, the probability of returning from 1 to 0 is $p/(1-p)$ in this last case, obtained by solving the two-boundary gambler's-ruin problem and sending the upper boundary to infinity. Repeated visits to zero restart the same experiment, so only finitely many occur. On the final excursion the driving increments obey the strong law and have positive mean.

Thus the three regimes are positive recurrence, null recurrence, and escape. Batch cleaning introduces further possibilities through the sizes of the downward jumps.

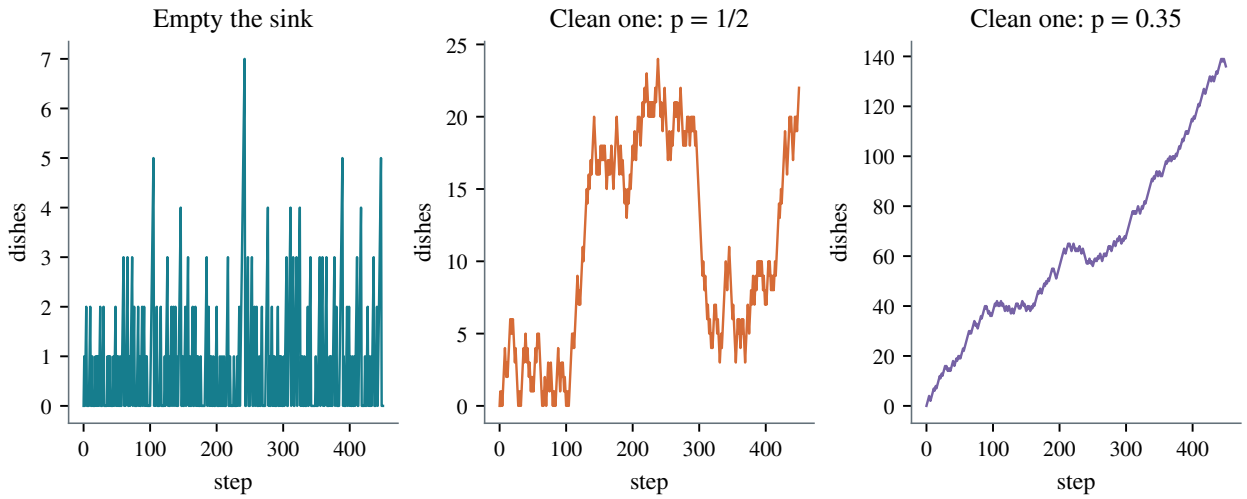


Figure 2: Three sample paths, each with 450 steps. Emptying the sink with probability 1/2 produces short recurrent excursions. Cleaning one dish with probability 1/2 gives a reflected symmetric walk; reducing that probability to 0.35 produces positive drift. The vertical scales are specific to each panel.

A continuous-time version can give additions rate 1 and cleaning rate $r(x)$. Its generator is

$$\mathcal{L}f(x) = f(x+1) - f(x) + r(x)[f(x-\beta(x)) - f(x)],$$

and its drift is $1 - r(x)\beta(x)$. Discrete-time probabilities and continuous-time rates therefore lead to different balance equations.

Question 2.2. For cleaning rules $\beta(x) = cx$ or cx^γ at large x, classify recurrence and escape under specified cleaning probabilities $p(x)$. When a stationary law exists, what is its tail, and what scaling describes long excursions? Specify the rule near zero as part of the model.

3 The coupon collector process

The first few coupons are easy to collect. The last few take most of the time: a large fraction of the album fills in order n draws, while completion takes order $n \log n$. Counting the missing types makes both scales visible.

Draw repeatedly and uniformly from $[n]$ with replacement. Let X_k count the types already seen and let $\tau = \min\{k : X_k = n\}$. The successive discovery waiting times are independent geometric random variables on $\{1, 2, \ldots\}$, with success probabilities $(n - i + 1)/n$ for discovery i. Hence

$$\mathbb{E}\tau = nH_n, \qquad \mathrm{Var}(\tau) = n^2 \sum_{j=1}^{n} j^{-2} - nH_n. \tag{3.1}$$

They become exponential waiting times after Poissonizing the draws. Directly, the union bound gives $\mathbb{P}(\tau > k) \le n(1 - 1/n)^k \le ne^{-k/n}$.

Write $a_k = (1 - 1/n)^k$ and $b_k = (1 - 2/n)^k$, with $n \ge 2$. Indicators that individual coupons are present give

$$\mathbb{E}X_k = n(1 - a_k), \tag{3.2}$$

$$\mathrm{Var}(X_k) = na_k(1 - a_k) + n(n - 1)(b_k - a_k^2). \tag{3.3}$$

For $k \le m$ the same calculation yields

$$\mathrm{Cov}(X_k, X_m) = na_m(1 - a_k) + n(n - 1)(b_k a_{m-k} - a_k a_m). \tag{3.4}$$

For example, the mixed probability that distinct coupons are both missing, the first at time k and the second at time m, is $b_k a_{m-k}$.

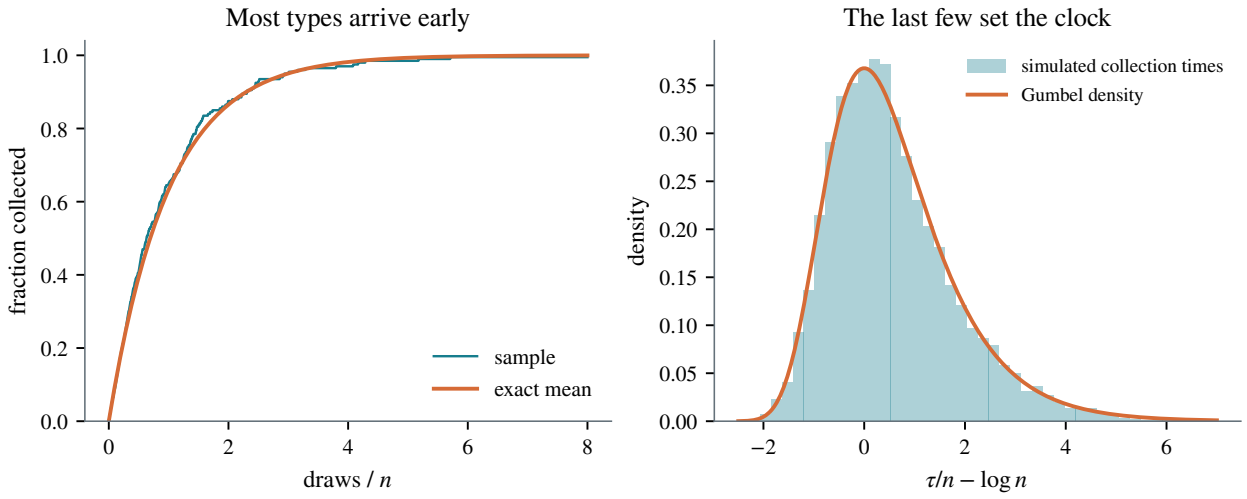


Figure 3: Left: one collection path with $n = 200$, together with its exact expected fraction collected. Right: 6000 independent completion times for $n = 1000$, generated by summing the independent geometric discovery waits. Centering at $n \log n$ and dividing by n reveals the Gumbel density.

The conditional expectation $\mathbb{E}(X_{k+1} \mid \mathcal{F}_k) = 1 + (1 - 1/n)X_k$ gives the martingale

$$M_k = X_k - k + \frac{1}{n} \sum_{j=0}^{k-1} X_j. \tag{3.5}$$

Taking expectations recovers the recursion for $\mathbb{E}X_k$. Stopping at τ gives

$$\mathbb{E}\sum_{j=0}^{\tau-1} X_j = n^2(H_n - 1). \tag{3.6}$$

To justify stopping, first use $\tau \wedge m$. The increments of M are bounded, and $\mathbb{E}\tau < \infty$, so the stopped variables converge in L^1. Similarly,

$$X_k^2 - k + \frac{2}{n}\sum_{j=0}^{k-1} X_j^2 - \left(2 - \frac{1}{n}\right)\sum_{j=0}^{k-1} X_j \tag{3.7}$$

is a martingale.

There are two useful time scales. For $k \sim nt$, (3.3) gives

$$\frac{\mathrm{Var}(X_k)}{n} \longrightarrow e^{-t} - (1+t)e^{-2t}.$$

For $k = \lfloor \alpha n \log n \rfloor$ with fixed $\alpha > 0$,

$$\mathbb{E}X_k = n(1 - n^{-\alpha}) + o(n^{(1-\alpha)/2}), \qquad \mathrm{Var}(X_k) \sim n^{1-\alpha}.$$

The growing variance for $0 < \alpha < 1$ suggests normal marginals, but the dependence across distinct values of α disappears.

Proposition 3.1. *For any fixed $0 < \alpha_1 < \cdots < \alpha_r < 1$, the vector*

$$\left(\frac{X_{\lfloor \alpha_j n \log n \rfloor} - n(1 - n^{-\alpha_j})}{n^{(1-\alpha_j)/2}}\right)_{j=1}^{r}$$

converges to r independent standard normal random variables.

Proof. Run the draws at rate one. The first arrival times of the n coupon types are then independent exponentials of rate $1/n$. At time $t_j = \alpha_j n \log n$, the missing-coupon indicators are independent across types and have probability $n^{-\alpha_j}$. Their standardized sums satisfy the multivariate triangular-array central limit theorem: each centered summand has norm tending uniformly to zero, and for $i < j$ the covariance of the standardized sums tends to zero, being asymptotic to $n^{-(\alpha_j - \alpha_i)/2}$.

Couple these draws to the original iid coupon sequence. Let $w_n = \lceil \sqrt{n}(\log n)^2 \rceil$. The Poisson count at each t_j differs from $\lfloor t_j \rfloor$ by at most w_n with probability tending to one. By monotonicity, the discrepancy in distinct coupons is then at most the number discovered between draws $\lfloor t_j \rfloor - w_n$ and $\lfloor t_j \rfloor + w_n$. Its expectation is $O(w_n n^{-\alpha_j})$, which is $o(n^{(1-\alpha_j)/2})$. Markov's inequality and the finite union bound complete the transfer. □

Distinct fixed values of α are separated by order $n \log n$ draws. At this resolution the normalized fluctuations lose their temporal memory. The final window has a different scale and a different law:

$$\mathbb{P}\left(\frac{\tau - n\log n}{n} \le x\right) \longrightarrow \exp(-e^{-x}). \tag{3.8}$$

To see this directly, Poissonize at rate one and let $\widetilde{\tau}$ be the maximum of the n independent first-arrival clocks. Then

$$\mathbb{P}(\widetilde{\tau} \le n\log n + nx) = (1 - n^{-1}e^{-x})^n \longrightarrow e^{-e^{-x}}.$$

Conditionally on the discrete collection time τ, the elapsed Poisson time is a sum of τ independent rate-one exponential waits, independent of the labels. Hence $\mathbb{E}(\widetilde{\tau} - \tau)^2 = \mathbb{E}\tau = nH_n = o(n^2)$, which transfers the limit. The Gaussian regime concerns many missing types; the Gumbel regime begins when only finitely many remain.

3.1 Generalized coupon collector

Now draw iid labels V_i with probabilities $(p_j)_{j\geq 1}$ and set $W_k = \{V_1, \dots, V_k\}$, $Z_k = |W_k|$, $W_0 = \varnothing$. Then

$$\mathbb{E}Z_k = \sum_j [1 - (1-p_j)^k]. \tag{3.9}$$

Its growth depends on the probability tail. Finite support gives a bounded limit; a geometric tail gives logarithmic growth; probabilities of order j^{-a} with $a > 1$ give order $k^{1/a}$. The last two orders follow by comparing each summand with $\min\{1, kp_j\}$ and splitting at $kp_j = 1$.

For a finite set A, let $g(A) = \sum_{j\notin A} p_j$, the probability of a *new* coupon. The compensated processes are

$$Z_k - \sum_{j=0}^{k-1} g(W_j), \qquad Z_k^2 - \sum_{j=0}^{k-1} (2Z_j + 1) g(W_j). \tag{3.10}$$

Both are martingales in the natural filtration. If T_r is the first time $Z_k = r$ and the support has at least r labels, then

$$\mathbb{E} \sum_{j=0}^{T_r - 1} g(W_j) = r.$$

Indeed, stop at $T_r \wedge m$, use $Z_{T_r \wedge m} \leq r$, and pass to the limit by bounded and monotone convergence. If S_m is the time all labels $1, \dots, m$ have appeared and their probabilities are positive, then $\mathbb{E}S_m \leq \sum_{j=1}^m p_j^{-1} < \infty$, so likewise $\mathbb{E}Z_{S_m} = \mathbb{E}\sum_{j=0}^{S_m - 1} g(W_j)$.

For two distinct labels a, b with positive probabilities, the first among them to appear has the elementary law

$$\mathbb{P}(\tau_a < \tau_b) = \sum_{k\geq 0} (1 - p_a - p_b)^k p_a = \frac{p_a}{p_a + p_b}. \tag{3.11}$$

This compares the two labels by ignoring all draws of other types.

Question 3.2. How well can the whole collected set, and its future trajectory, be approximated by independent occupancy clocks? What changes when observation stops at a discovery time?

Poissonization assigns label j an independent first-arrival clock of rate p_j. This gives an exact independent construction in elapsed time. The subsequent task is to control the random change back to a fixed number of draws, especially near discovery times selected by the process itself.

4 Non-adjacent uniform placements

Fill a row of n seats by repeatedly choosing a uniformly random available seat and blocking its neighbors. The final density is about 0.4323. There are two useful ways to obtain this constant: split the row at its first arrival, or ask whether one site survives the competition from its neighbors. The first gives generating functions; the second reveals the spatial law.

Let X_n be the final number seated. Then $\lceil n/3 \rceil \le X_n \le \lceil n/2 \rceil$. The first occupied seat, say U, deletes itself and its neighbors, leaving two independent instances on the remaining intervals. With $X_{-1} = X_0 = 0$,

$$X_n \stackrel{d}{=} 1 + X'_{U-2} + X''_{n-U-1}, \qquad U \sim \mathrm{Unif}[n], \tag{4.1}$$

where the two arrays on the right are independent of each other and of U. One way to see this independence is to assign iid continuous priorities to the seats and examine them in increasing priority order, accepting a seat when neither neighbor has been accepted. Conditional on the first seat, the relative priority orders in the two remaining intervals are independent.

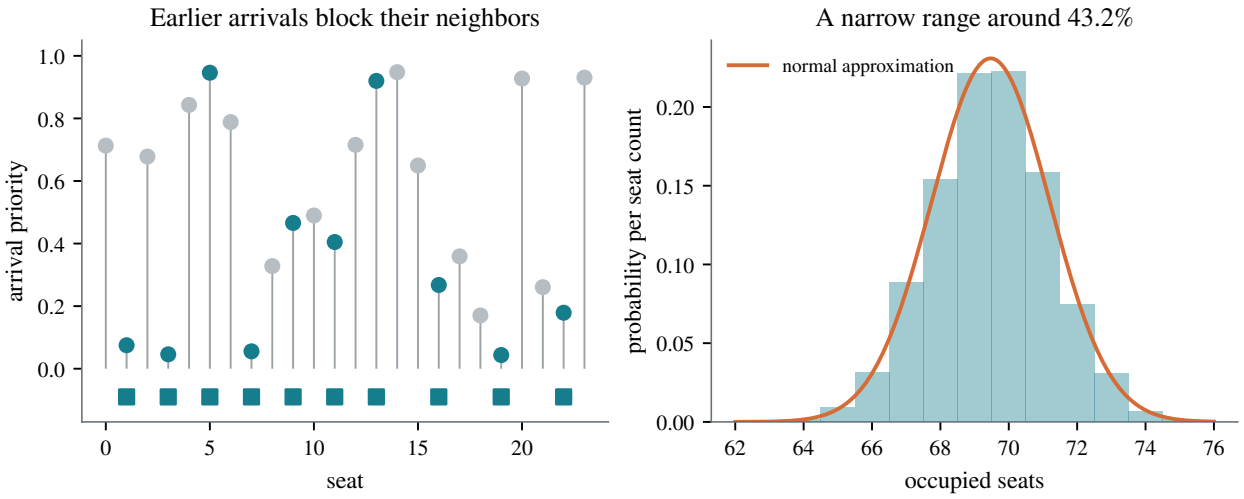


Figure 4: Left: independent arrival priorities on 24 seats, with accepted arrivals in teal and their final positions marked below the axis. Right: the final number occupied in 5000 independent rows of length 160, compared with the normal approximation using the displayed mean and variance asymptotics.

For $a_n = \mathbb{E}X_n$ and $f(x) = \sum_{n\ge1} a_n x^n$, the recursion gives

$$a_n = 1 + \frac{2}{n}\sum_{i=1}^{n-2} a_i, \qquad f'(x) = \frac{1}{(1-x)^2} + \frac{2x}{1-x} f(x).$$

Solving with $f(0) = 0$ yields

$$f(x) = \frac{1 - e^{-2x}}{2(1-x)^2}, \qquad a_n = \frac{1}{2}\sum_{j=1}^{n} \frac{(-1)^{j+1}2^j}{j!}(n+1-j). \tag{4.2}$$

The exponential series then gives

$$a_n = \frac{1-e^{-2}}{2} n + \frac{1-3e^{-2}}{2} + O(e^{-n}). \tag{4.3}$$

The actual remainder decreases faster than exponentially; the displayed bound is a convenient weaker form.

Squaring (4.1), using independence, and writing $b_n = \mathbb{E}X_n^2$, $g(x) = \sum_{n\ge1} b_n x^n$, gives

$$nb_n = 2na_n - n + 2\sum_{i=1}^{n-4} a_i a_{n-3-i} + 2\sum_{i=1}^{n-2} b_i, \tag{4.4}$$

$$g'(x) = 2f'(x) - \frac{1}{(1-x)^2} + 2x^2 f(x)^2 + \frac{2x}{1-x} g(x), \tag{4.5}$$

$$g(x) = \frac{(1+x)(1+e^{-4x}) - 2(1-x+2x^2)e^{-2x}}{4(1-x)^3}. \tag{4.6}$$

Expanding at $x = 1$ and subtracting a_n^2 gives

$$\operatorname{Var}(X_n) = e^{-4}(n+3) + O(e^{-n}). \tag{4.7}$$

In particular, $X_n/n \to (1-e^{-2})/2$ in L^2.

The full distribution has a similarly useful generating function. Set $c_n(z) = \mathbb{E}z^{X_n}$, with $c_{-1} = c_0 = 1$, and $h(x,z) = \sum_{n\ge1} c_n(z)x^n$. Then

$$\partial_x h = z(1+x+xh)^2, \qquad h(0,z) = 0, \tag{4.8}$$

and its solution, for $z > 0$ near 1 and x near 0, is

$$h(x,z) = -\frac{x\sqrt{z}\cosh(x\sqrt{z}) + (xz+z-1)\sinh(x\sqrt{z})}{(x-1)\sqrt{z}\cosh(x\sqrt{z}) + (xz-1)\sinh(x\sqrt{z})}. \tag{4.9}$$

It can also be read as a power-series identity at $x = 0$.

4.1 The normal limit and what the pole tells us

This is one-dimensional blocking random sequential adsorption. Penrose and Sudbury [4, Theorem 11] prove the following more general statement. Give the sites independent uniform arrival times in $(0,1)$ and let $X_n(t)$ count accepted sites by time t.

Theorem 4.1 (Penrose–Sudbury: one-dimensional blocking)**.** *For fixed $0 < t \le 1$, on a finite path of n sites,*

$$\frac{\operatorname{Var} X_n(t)}{n} \longrightarrow te^{-4t}, \qquad \frac{X_n(t) - \mathbb{E}X_n(t)}{\sqrt{n}} \longrightarrow_d N(0, te^{-4t}).$$

The same conclusions hold when counting n consecutive sites of the infinite-line process.

The normalization is $\sqrt{n}$, as in their Theorem 10; the displayed n^{-1} in (21)–(22) of the arXiv reprint is a typographical error. Their proof uses the rapidly decreasing probability of a long dependency path. The next argument proves the terminal-time conjecture directly from the generating function in this notebook.

Proposition 4.2. *Put $\mu = (1-e^{-2})/2$ and $v = e^{-4}$. Then*

$$\frac{X_n - a_n}{\sqrt{\operatorname{Var} X_n}} \longrightarrow_d N(0,1), \qquad \sup_x \left| \mathbb{P}\left(\frac{X_n - a_n}{\sqrt{\operatorname{Var} X_n}} \le x \right) - \Phi(x) \right| = O(n^{-1/2}).$$

Moreover, the third centered moment satisfies

$$\mathbb{E}(X_n - a_n)^3 = \frac{e^{-2} - 61e^{-6}}{16}\, n + O(1). \tag{4.10}$$

Proof. Write $D(x,z)$ for the denominator in (4.9). At $z=1$, $D(x,1)=(x-1)e^x$. Fix $R>1$. The implicit-function theorem and Rouché's theorem give, for complex z sufficiently close to one, a unique zero $\rho(z)$ in $|x|<R$, simple and analytic, with $\rho(1)=1$. There is no cancellation at this pole. Subtract its principal part and apply Cauchy's coefficient estimate on $|x|=R$ to obtain, uniformly there,

$$\mathbb{E}z^{X_n} = A(z)\rho(z)^{-n} + O(R^{-n}), \qquad A(1)=1, \tag{4.11}$$

with A analytic and nonzero near one. After shrinking the neighborhood, the error is exponentially small relative to the first term. Hence

$$\log \mathbb{E}e^{sX_n} = n\lambda(s) + b(s) + O(q^n), \quad \lambda(s) = -\log\rho(e^s), \quad b(s) = \log A(e^s), \quad q<1,$$

locally uniformly for complex s; Cauchy's formula also controls derivatives.

Set $y(s) = e^{s/2}\rho(e^s)$. The pole equation becomes

$$(y-1)e^{2y} = \tanh(s/4)(y+1), \qquad y(0)=1.$$

Implicit differentiation gives

$$y'(0) = \tfrac{1}{2}e^{-2}, \quad y''(0) = -\tfrac{3}{4}e^{-4}, \quad y'''(0) = \tfrac{39}{16}e^{-6} - \tfrac{1}{16}e^{-2}.$$

Since $\lambda(s) = s/2 - \log y(s)$,

$$\lambda'(0) = \mu, \quad \lambda''(0) = v, \quad \lambda'''(0) = (e^{-2} - 61e^{-6})/16.$$

The derivatives of the logarithmic moment generating function are the cumulants, proving (4.10) and the mean and variance rates. Taylor expansion at $s = t/\sqrt{\operatorname{Var} X_n}$ proves convergence of the centered moment generating functions to $e^{t^2/2}$.

For completeness, the same expansion gives the rate. If φ_n is the characteristic function of the standardized variable, analyticity and $v>0$ give constants $c, C, \delta > 0$ with

$$|\varphi_n(t) - e^{-t^2/2}| \le Cn^{-1/2}|t|^3 e^{-ct^2}, \qquad |t| \le \delta\sqrt{n}.$$

Indeed the remainder in its logarithm is $O(|t|^3/\sqrt{n})$; choosing δ small makes its real part at most a fixed fraction of t^2. The smoothing inequality

$$\sup_x |F_n(x) - \Phi(x)| \le C\int_{-T}^{T} \left| \frac{\varphi_n(t) - e^{-t^2/2}}{t} \right| dt + C/T$$

with $T = \delta\sqrt{n}$ proves the bound. This is the analytic quasi-power mechanism developed systematically by Hwang [5]. □

In particular, the leading standardized skewness is $(e^4 - 61)/(16\sqrt{n}) \approx -0.4001/\sqrt{n}$. The same pole controls all fixed cumulants, and its location for real s also gives exponential Chernoff bounds. This supplies information beyond the original normal-limit question without changing the model.

4.2 The infinite-line construction

Assign iid uniform priorities W_z to $z \in \mathbb{Z}$, and accept each site at its arrival if neither neighbor has already been accepted. This defines a random maximal independent set I. On an infinite line the construction is justified by decreasing priority paths: a dependency path of r edges in either

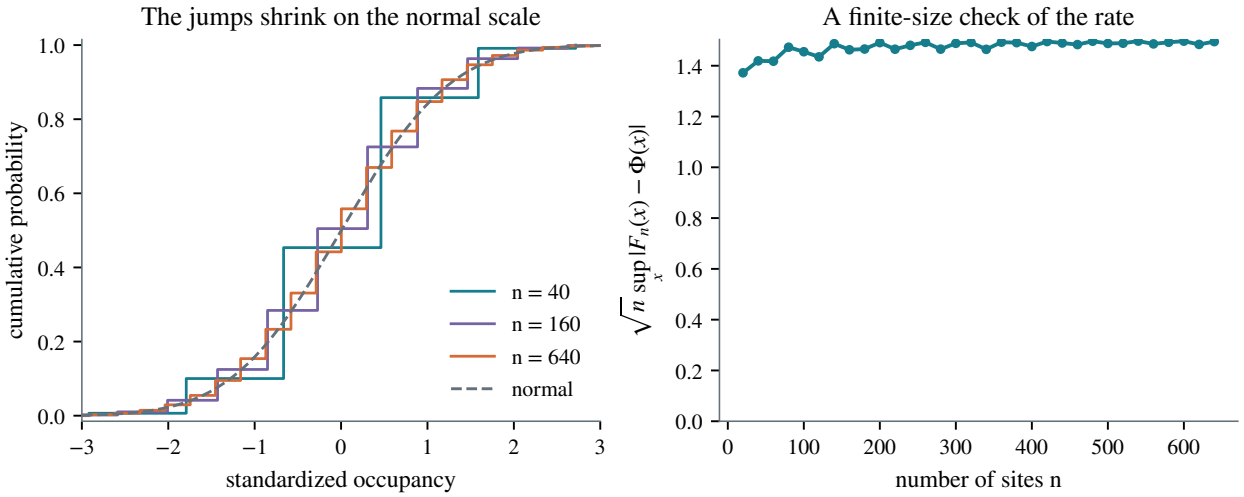


Figure 5: Exact terminal occupancy distributions from the splitting recursion. Left: standardized distribution functions for three path lengths. Right: the Kolmogorov distance multiplied by $\sqrt{n}$ for the computed lengths. The comparison includes both sides of every jump in the discrete CDF.

fixed direction has probability $1/(r+1)!$. There is almost surely no infinite decreasing path, so each acceptance decision depends on finitely many priorities.

Let $q_n(x)$ be the probability that the endpoint of an n-site path has not been accepted before time x. Set $q_0(x) = 1$. Conditioning on its arrival, the endpoint is accepted at time u exactly when its neighbor in the remaining path is still unaccepted. Thus

$$q_n(x) = 1 - \int_0^x q_{n-1}(u)\,du = \sum_{j=0}^{n} \frac{(-x)^j}{j!}.$$

In the half-line limit $q(x) = e^{-x}$. Before site 0 arrives, the left and right half-lines evolve independently. Therefore

$$\mathbb{P}(0 \in I) = \int_0^1 e^{-2x}\,dx = \frac{1 - e^{-2}}{2}. \tag{4.12}$$

This explains the constant in a_n/n directly. On the half-line $\{0, 1, \ldots\}$ the endpoint is accepted with probability $1 - e^{-1}$. For a finite interval $[-a, b]$,

$$\mathbb{P}(0 \in I([-a, b])) = \int_0^1 q_a(x) q_b(x)\,dx. \tag{4.13}$$

Convergence to the infinite-line value requires both a and b to tend to infinity.

4.3 Distribution of the configuration

Between consecutive occupied sites there are either one or two empty sites. It is tempting to model those gap sizes as independent. That renewal model has a different distribution. Conditional on the priorities at four alternating sites, the true construction gives

$$\begin{aligned}\mathbb{P}(0, 2, 4, 6 \in I \mid W_0 = x, W_2 = y, W_4 = z, W_6 = w) \\ = e^{-x-w}(1 - \min(x, y))(1 - \min(y, z))(1 - \min(z, w)).\end{aligned} \tag{4.14}$$

The exterior half-lines give e^{-x-w}; each intervening site must arrive after at least one of its two accepted neighbors. Integrating over the unit four-cube gives

$$\mathbb{P}(0, 2, 4, 6 \in I) = \frac{1}{2} - \frac{79}{30e^2} \approx 0.1436171. \tag{4.15}$$

For a stationary renewal process R with the same density, the probabilities of one and two empty sites per gap would be

$$p_1 = \frac{e^2 - 3}{e^2 - 1}, \qquad p_2 = \frac{2}{e^2 - 1}.$$

It would instead give $\mathbb{P}(0, 2, 4, 6 \in R) = \frac{1}{2}(1 - e^{-2})p_1^3 \approx 0.1401590$.

Decreasing runs are still useful for local exploration. Define

$$K = \min\{k \geq 0 : W_k < W_{k+1}\}.$$

Then

$$\mathbb{P}(K \geq k) = \frac{1}{(k+1)!}, \qquad \mathbb{P}(K = k) = \frac{k+1}{(k+2)!}, \tag{4.16}$$

and

$$\mathbb{E}K = e - 2, \quad \operatorname{Var} K = e(3 - e), \quad \mathbb{E}w^K = \frac{1 + (w-1)e^w}{w^2}, \tag{4.17}$$

with the removable singularity at $w = 0$ filled in. Exploring such runs locates the finite region of priorities that determines a site's decision. The accepted configuration still couples neighboring explorations through their common sites. The useful consequence is quantitative localization.

The factorial tail does imply rapid decay of correlations. Decide acceptance at 0 and d using only disjoint radius-r neighborhoods, with $2r < d$. The two truncated decisions are independent. Each differs from the true decision only if a decreasing path reaches the neighborhood boundary, an event of probability at most $2/(r+1)!$. Comparing products and means gives

$$|\operatorname{Cov}(1\{0 \in I\}, 1\{d \in I\})| \leq \frac{8}{(r+1)!}. \tag{4.18}$$

Question 4.3. Compare the local pattern probabilities of this greedy law with those of a uniformly chosen maximal independent set on a long interval. Which patterns most strongly distinguish the two measures?

Exact pair correlations for blocking adsorption are also available in [4]. They provide a useful benchmark for a second direction: improving the density under a specified restriction on local information. Coordinated alternating seats attain $1/2$; the interesting optimization is the cost of making decisions locally.

5 Adjacent occurrences in a random permutation

A random permutation occasionally places consecutive values in consecutive positions and in increasing order. There are $n-1$ possible positions for such a succession, each with probability $1/n$. More surprisingly, every prescribed set of successions can be counted by the same block argument.

Let σ be a uniform permutation of $[n]$ and set

$$A_j = 1\{\sigma(j+1) = \sigma(j)+1\}, \quad 1 \le j < n, \qquad Y_n = \sum_{j=1}^{n-1} A_j. \tag{5.1}$$

For $\sigma = 2341$, this gives $A_1 = A_2 = 1$, $A_3 = 0$. Both positions and values are linear here: the last and first positions are not joined, and $n, 1$ are not consecutive values.

Proposition 5.1. *For any $J \subseteq \{1, \dots, n-1\}$ with $|J| = r$,*

$$\mathbb{P}(A_j = 1 \text{ for all } j \in J) = \frac{(n-r)!}{n!}. \tag{5.2}$$

Consequently, for $n \ge 2$,

$$\mathbb{E}Y_n = 1 - \frac{1}{n}, \qquad \mathrm{Var}(Y_n) = 1 - \frac{1}{n} - \frac{1}{n^2}. \tag{5.3}$$

Proof. Join position j to $j+1$ whenever $j \in J$. This produces $n-r$ distinguishable blocks, each of which must receive consecutive increasing values. Order the blocks by their smallest values; any ordering of these $n-r$ blocks uniquely assigns the intervals of values, and every admissible permutation arises this way. Thus there are $(n-r)!$ admissible permutations. In particular $\mathbb{E}A_j = 1/n$ and $\mathbb{E}A_iA_j = 1/[n(n-1)]$ for $i \ne j$. Summing yields the stated moments. □

The intersection probability therefore depends only on the number of specified adjacencies. For distinct i, j, the covariance is $1/[n^2(n-1)]$.

Theorem 5.2. *Y_n converges in distribution to* Poi(1).

Proof. The factorial moments are

$$\mathbb{E}(Y_n)_r = (n-1)_r \frac{(n-r)!}{n!} = 1 - \frac{r}{n}, \qquad 0 \le r \le n-1.$$

For $z \ge 0$, expanding $z^{Y_n} = (1 + (z-1))^{Y_n}$ gives

$$\mathbb{E}z^{Y_n} = \sum_{r=0}^{n-1} \Big(1 - \frac{r}{n}\Big) \frac{(z-1)^r}{r!} \longrightarrow e^{z-1}.$$

The exponential series dominates absolutely, including in a neighborhood of $z = 1$. Taking $z = e^t$ proves convergence by moment-generating functions. □

Extracting coefficients gives the full finite distribution:

$$\mathbb{P}(Y_n = j) = \frac{1}{j!} \sum_{r=j}^{n-1} \Big(1 - \frac{r}{n}\Big) \frac{(-1)^{r-j}}{(r-j)!}, \qquad 0 \le j \le n-1. \tag{5.4}$$

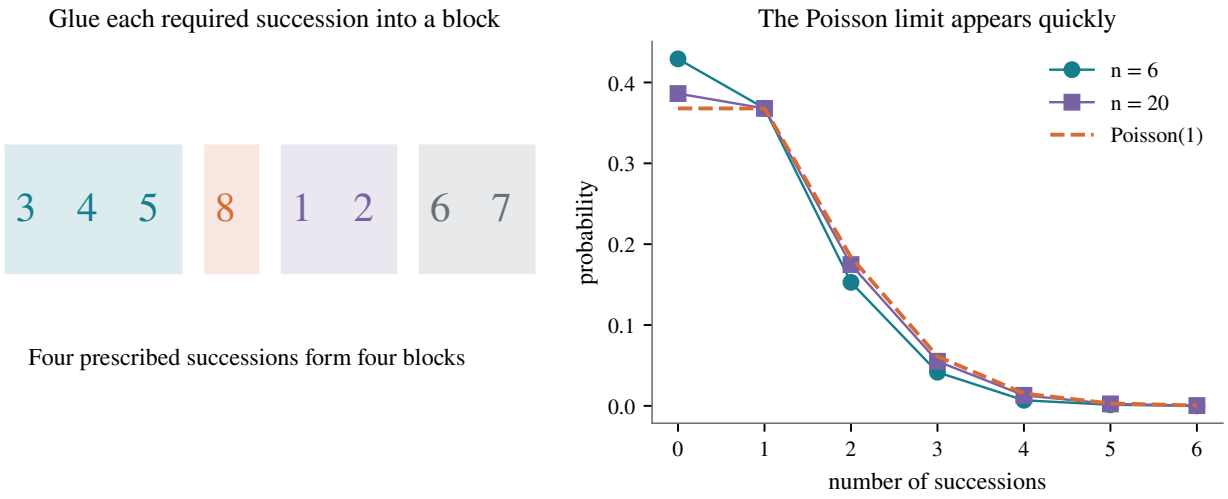


Figure 6: Successions group the values into increasing blocks. The example has eight values and four successions, leaving four blocks. The right panel compares the exact mass function (5.4) for $n = 6, 20$ with Poisson(1).

Thus the same calculation gives exact probabilities, moments, and their Poisson limit. It also explains why the approximation is already effective for modest n.

There is also a natural coupling in n: insert $n+1$ at one of the $n+1$ possible positions, uniformly. Conditional on the present permutation,

$$Y_{n+1} = \begin{cases} Y_n - 1, & \text{with probability } Y_n/(n+1), \\ Y_n, & \text{with probability } (n - Y_n)/(n+1), \\ Y_n + 1, & \text{with probability } 1/(n+1). \end{cases} \tag{5.5}$$

The decreasing moves break an existing adjacency, and the increasing move places $n+1$ just after n. The factor $1/(n+1)$ slows this evolution as the permutation grows. Factorial moments avoid that changing time scale and identify the limit directly.

Joining the last position to the first adds a single indicator with mean $1/n$. Allowing the additional value adjacency $n \to 1$ around that cycle adds another indicator, with mean $1/(n-1)$ for $n \geq 2$. Both changes tend to zero in L^1, so all three adjacency counts have the same Poisson limit.

6 Hitting times of sums

Adding increments and inverting the resulting staircase are two descriptions of the same experiment. The inverse can have a simpler law: geometric increments turn into a binomial count, while uniform increments produce a delay equation with an explicit solution.

6.1 IID uniform increments

Let U_i be iid uniform on $(0,1)$, $S_0 = 0$, $S_n = \sum_{i=1}^{n} U_i$, and $\tau_x = \min\{n \geq 1 : S_n \geq x\}$ for $x > 0$. Set $m(x) = \mathbb{E}\tau_x$. The tail-sum formula is

$$m(x) = \sum_{k \geq 0} \mathbb{P}(S_k < x). \tag{6.1}$$

For x in a bounded interval $[0, A]$, exponential Markov bounds give $\mathbb{P}(S_k < x) \leq e^{sA}(\mathbb{E}e^{-sU_1})^k$ for any $s > 0$. Thus the tails are uniformly summable. In particular, m is continuous on $(0, \infty)$.

Conditioning on the first increment gives

$$m(x) = 1 + \int_0^{\min(x,1)} m(x-u)\, du. \tag{6.2}$$

For $0 < x \leq 1$, this implies $m'(x) = m(x)$ and $m(0+) = 1$, hence $m(x) = e^x$. For $x > 1$, it gives $m'(x) = m(x) - m(x-1)$. Solving successively on unit intervals,

$$m(x) = \sum_{j=0}^{\lfloor x \rfloor} \frac{(-1)^j}{j!}(x-j)^j e^{x-j}. \tag{6.3}$$

At a positive integer $x = j$, the new summand vanishes; the $j = 0$ term is always e^x. For example, $m(x) = e^x - (x-1)e^{x-1}$ on $1 < x \leq 2$. At the integer $j \geq 1$, m is C^{j-1} and its jth derivative has a jump.

The same calculation gives an especially short second-moment formula. For fixed $x > 0$ define

$$H(z, x) = \sum_{k \geq 0} z^k \mathbb{P}(S_k < x).$$

Inclusion–exclusion for the uniform-sum distribution gives

$$H(z, x) = \sum_{j=0}^{\lfloor x \rfloor} \frac{(-1)^j [z(x-j)]^j}{j!} e^{z(x-j)}. \tag{6.4}$$

To obtain this, use $\mathbb{P}(S_k < x) = \frac{1}{k!} \sum_{j=0}^{\min(k, \lfloor x \rfloor)} (-1)^j \binom{k}{j} (x-j)^k$ for $k \geq 1$ and sum the exponential series. For every fixed positive z the sum over k converges, by choosing s large enough in the exponential bound above. Since $\mathbb{E}z^{\tau_x} = 1 + (z-1)H(z,x)$, differentiation at $z = 1$ gives

$$\mathbb{E}\tau_x^2 = (2x+1)m(x), \qquad \mathrm{Var}(\tau_x) = m(x)[2x + 1 - m(x)]. \tag{6.5}$$

Indeed, every summand in (6.4) has derivative at $z = 1$ equal to x times that summand. Thus the variance is explicit for every $x > 0$.

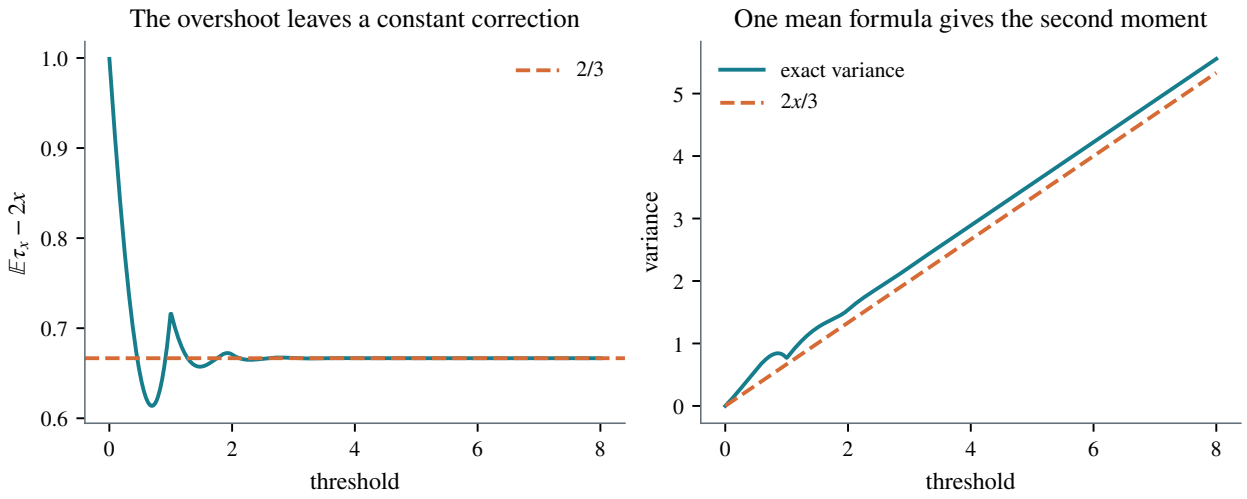


Figure 7: Uniform increments. The mean excess over $2x$ settles at $2/3$, with the first integer thresholds still visible. The exact variance follows from the same mean through (6.5); its leading slope is $2/3$.

For $0 < x \le 1$, higher moments can be written using Stirling numbers of the second kind:

$$\mathbb{E}\tau_x^k = e^x \sum_{j=1}^{k} j \left\{ {k \atop j} \right\} x^{j-1}. \tag{6.6}$$

This follows by applying $(z\partial_z)^k$ to $1 + (z-1)e^{zx}$ at $z = 1$. The first three are e^x, $(1+2x)e^x$, and $(1+6x+3x^2)e^x$. On $(0,1)$, the variance has a maximum at the unique root of $3 + 2x = 2e^x$, approximately 0.858.

For large x, Wald's identity and $x \le S_{\tau_x} < x + 1$ already show $2x \le m(x) < 2x + 2$. The nonlattice renewal theorem for the excess [6, Section 2.6] gives

$$m(x) = 2x + \frac{2}{3} + o(1), \qquad \mathrm{Var}(\tau_x) \sim \frac{2}{3}x. \tag{6.7}$$

In detail, the limiting excess has mean $\mathbb{E}U_1^2/(2\mathbb{E}U_1) = 1/3$; bounded excesses make this convergence valid in expectation, and Wald multiplies that mean by two. Formula (6.5) then gives the variance asymptotic. Inverting the ordinary central limit theorem for S_n also yields

$$\frac{\tau_x - 2x}{\sqrt{(2/3)x}} \;\longrightarrow_d\; N(0,1).$$

For instance, at $n = \lfloor 2x + y\sqrt{(2/3)x} \rfloor$, $\mathbb{P}(\tau_x \le n) = \mathbb{P}(S_n \ge x) \to \Phi(y)$.

An explicit exponential remainder

The residual distance to the next renewal provides a short quantitative proof of the $2/3$ correction. It also proves the exponential convergence suggested by the exact formula.

Proposition 6.1. *For $x > 0$,*

$$|m(x) - 2x - 2/3| \le 2^{1-\lfloor x \rfloor}, \qquad \mathrm{Var}(\tau_x) = \frac{2}{3}x + \frac{2}{9} + O((x+1)2^{-x}). \tag{6.8}$$

Proof. Let $R_x = S_{\tau_x} - x$ be the residual lifetime. At $x = 0$ use the first strictly positive renewal, so $R_0 = U_1$. Viewed as a process in the level x, R decreases at speed one and restarts with an independent uniform lifetime whenever it reaches zero.

Starting at residual $r \in [0, 1)$, one unit of level leaves a renewal problem at threshold $a = 1 - r \in (0, 1]$. On this interval the renewal density is $\sum_{j\ge1} f_{S_j}(z) = e^z$, $0 < z < 1$. Including the renewal at zero, the density of the excess $y \in (0, 1)$ is

$$p_a(y) = \begin{cases} e^a, & a + y \le 1, \\ e^a - e^{a+y-1}, & a + y > 1. \end{cases}$$

In the second case it is at least $e^{1-y} - 1 \ge 1 - y$; the first case has the same lower bound. Thus the one-unit transition kernel P satisfies

$$P(r, dy) \ge (1 - y)\, dy = \tfrac{1}{2}\pi(dy), \qquad \pi(dy) = 2(1 - y)\, dy.$$

Direct integration verifies $\int_0^1 p_{1-r}(y) 2(1 - r)\, dr = 2(1 - y)$, so $\pi P = \pi$. Writing $P = \frac{1}{2}\pi + \frac{1}{2}Q$ shows that every probability law is within 2^{-k} of π in total variation after k steps. For $x = k + s$, $0 \le s < 1$, apply this bound to the initial law of R_s. Since $0 \le y \le 1$ and $\int y\, \pi(dy) = 1/3$,

$$|\mathbb{E}R_x - 1/3| \le 2^{-\lfloor x \rfloor}.$$

Wald's identity gives $m(x) = 2x + 2\mathbb{E}R_x$. Substitution in $\mathrm{Var}(\tau_x) = m(x)[2x + 1 - m(x)]$ proves the variance expansion. □

The proof uses a uniform chance of forgetting the current excess during each unit of level. It gives an explicit rate without locating complex zeros. The transform

$$\int_0^\infty e^{-sx} m(x)\, dx = (s - 1 + e^{-s})^{-1} = 2s^{-2} + \tfrac{2}{3}s^{-1} + O(1)$$

suggests a sharper description through its nonzero poles.

Question 6.2. What is the leading exponentially damped oscillation of $m(x) - 2x - 2/3$? Can its amplitude and phase be expressed cleanly using the nearest nonzero roots of $s - 1 + e^{-s}$?

6.2 IID geometric increments

Let the increments now be geometric with $\mathbb{P}(X_i = j) = p(1 - p)^{j-1}$ for $j \ge 1$, and set $\sigma_k = \min\{s \ge 0 : S_s \ge k\}$ for positive integers k. Real thresholds reduce to $\lceil k \rceil$ because the sums are integer-valued.

Realize the X_i as successive waiting times between successes in an iid Bernoulli(p) sequence. There are $\mathrm{Bin}(k - 1, p)$ successes strictly before time k. The first success at or after k has the next index, so

$$\sigma_k \overset{d}{=} 1 + \mathrm{Bin}(k - 1, p), \qquad \mathbb{E}\sigma_k = 1 + (k - 1)p, \qquad \mathrm{Var}(\sigma_k) = (k - 1)p(1 - p). \tag{6.9}$$

This also explains the generating function $\sum_{k\ge1} z^k \mathbb{E}\sigma_k = z(1 - (1 - p)z)/(1 - z)^2$. The inverse relation with the partial sums is $\{\sigma_k > s\} = \{S_s < k\}$.

If an observer succeeds only when the hitting index lies in a block of $L + 1$ consecutive positions, the optimal placement is a binomial window problem. If $b_r = \binom{k-1}{r} p^r (1 - p)^{k-1-r}$, set $b_r = 0$ outside the support. The chance that $\sigma_k \in \{j - L, \dots, j\}$ is $\sum_{r=j-L-1}^{j-1} b_r$. Moving the window one place right changes this probability by $b_j - b_{j-L-1}$. Thus the best positions and the medians can be found directly from the binomial probabilities, including any ties.

7 Heatseekers

Cyclic pursuit continually changes the directions of motion: each particle chases the next, and a collision shortens the cycle. In Euclidean space a single scalar quantity, the polygonal perimeter, controls the entire configuration. On a circle, translating configurations can persist.

Start k heatseeking missiles in $\mathbb{R}^d$, labelled cyclically. Each moves at unit speed toward the next missile. When a missile reaches its target, merge that consecutive pair and continue along the shortened cycle. Simultaneous zero-length edges are removed together. In Euclidean space every finite configuration coalesces completely in finite time.

Proposition 7.1. *If the initial polygonal perimeter is L_0 and there are $k \geq 2$ missiles, complete coalescence occurs by time kL_0.*

Proof. Between collisions, write $e_i = x_{i+1} - x_i$, $u_i = e_i/|e_i|$, and $L = \sum_i |e_i|$. Then

$$L' = \sum_i \langle u_i, u_{i+1} - u_i \rangle = -\tfrac{1}{2} \sum_i |u_{i+1} - u_i|^2.$$

For any chosen u_i, the identity $\sum_j |e_j| u_j = 0$ implies that some u_j has $\langle u_i, u_j \rangle \leq 0$. Hence $|u_i - u_j|^2 \geq 2$. Along a path of at most k edges in the cyclic list, Cauchy–Schwarz gives $\sum_i |u_{i+1} - u_i|^2 \geq 2/k$, so $L' \leq -1/k$. At a collision, deleting zero-length edges preserves L. The number of missiles decreases, so the bound with the original k remains valid. There are finitely many mergers; as long as at least two distinct missiles remain, the nonnegative perimeter decreases at rate at least $1/k$. □

For a regular k-gon, the whole motion is explicit. If $r(t)$ is its circumradius and $\varphi(t)$ its rotation angle, with $\vartheta = \pi/k$, then

$$r(t) = r_0 - t \sin \vartheta, \qquad \varphi(t) - \varphi(0) = \cot \vartheta \log \frac{r_0}{r(t)}.$$

Indeed, the unit velocity toward the next vertex has radial component $-\sin\vartheta$ and tangential component $\cos\vartheta$. The particles therefore trace logarithmic spirals and meet at time $r_0/\sin\vartheta$. Finite-time collapse is a classical result in cyclic pursuit; Gekhtman [7] develops the connection with geometry on compact manifolds.

On a flat torus, specify that motion follows a shortest displacement, with a fixed rule for ties. Traveling configurations can persist. On the circle, if all directed shortest displacements have the same sign, every missile moves in the same direction and all separations stay constant.

These initial signs are dependent. To compute the probability of this event for iid uniform initial positions on the circle, let $d_i \in (0,1)$ be the clockwise displacement from missile i to $i+1$. The first $k-1$ displacements are independent uniforms, and $d_k = (-\sum_{i<k} d_i) \bmod 1$. Therefore

$$\mathbb{P}(\text{all shortest directions agree}) = 2 \int_{(0,1/2)^{k-1}} 1\left\{ \left\{ \sum_{i=1}^{k-1} u_i \right\} > \frac{1}{2} \right\} du_1 \cdots du_{k-1}, \tag{7.1}$$

where inner braces denote fractional part. For $k = 2$ this probability is zero, and for $k = 3$ it is $1/4$. A product of independent sign probabilities would miss the cyclic constraint.

Question 7.2. On a torus, what is the probability of eventual coalescence from iid initial points? What are the surviving cluster sizes and the distribution of locations after a specified number of coalescences? For Euclidean initial points, what scaling describes coalescence times and locations as $k \to \infty$?

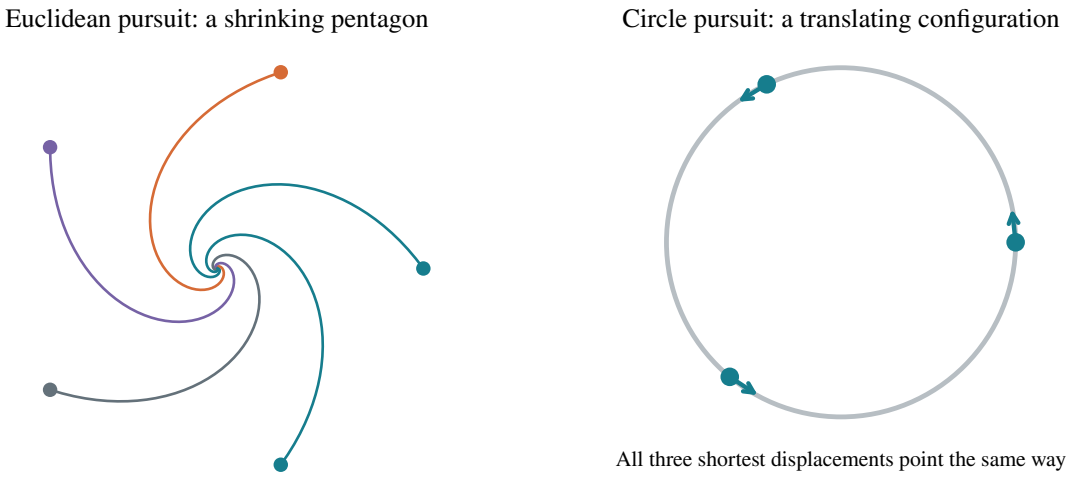


Figure 8: Left: exact logarithmic-spiral trajectories from the vertices of a regular pentagon. Right: three points at circle coordinates $0, 0.32, 0.64$. All three shortest directed displacements point counterclockwise, so this configuration translates with constant separations.

7.1 Mean field arrows

Start with iid fair signs on a cyclic list of length k. Choose uniformly among disagreeing directed edges and delete the first sign of the chosen edge. As long as both signs occur, there are equally many $+-$ and $-+$ edges. Consequently, the numbers (X_t, Y_t) of each sign form the walk that subtracts $(1,0)$ or $(0,1)$ with equal probability, starting at $(B, k-B)$ with $B \sim \mathrm{Bin}(k, 1/2)$ and stopping at

$$T = \min\{t : X_t Y_t = 0\}.$$

Write $R = k - T$ for the number of surviving arrows.

Proposition 7.3. *For this initialization,*

$$\mathbb{E}(R^2) + \mathbb{E}R = 2k, \qquad \frac{R}{\sqrt{k}} \longrightarrow_d |N(0,2)|, \qquad \mathbb{E}T = k - 2\sqrt{k/\pi} + o(\sqrt{k}). \tag{7.2}$$

Proof. The difference $D_t = X_t - Y_t$ is a simple symmetric random walk up to T, with $\mathbb{E}D_0^2 = k$. Since $T \le k$, the stopped square martingale gives $\mathbb{E}D_T^2 = k + \mathbb{E}T$. At absorption $|D_T| = R$, proving the first identity. In particular, $\mathbb{E}R \le \sqrt{2k}$ and $T/k \to 1$ in probability. Extend the coin increments to time k. Conditional on the information at T, their remaining sum has mean zero and variance $k - T = R$, so $\mathbb{E}(D_k - D_T)^2 = \mathbb{E}R = O(\sqrt{k})$. Meanwhile D_k is the sum of $2k$ independent fair signs, counting the k initial signs and the k walk increments. The central limit theorem gives $D_k/\sqrt{k} \to_d N(0,2)$, and thus the assertion for R. The first identity bounds $\mathbb{E}(R/\sqrt{k})^2$, giving uniform integrability and convergence of its mean to $2/\sqrt{\pi}$. □

Let Z_t count disagreeing edges. In a mixed configuration it is also the number of maximal constant-sign runs. Deleting from a run of length at least two leaves Z_t unchanged. Deleting a singleton run merges its neighbors and decreases Z_t by two, including the final transition to a constant configuration. A constant configuration has $Z_t = 0$.

Let $N_j(t)$ be the number of runs of length j, before absorption. Then

$$Z_t = \sum_{j \ge 1} N_j(t), \qquad k - t = \sum_{j \ge 1} j N_j(t), \qquad \mathbb{E}(Z_{t+1} - Z_t \mid \mathcal{F}_t) = -2\frac{N_1(t)}{Z_t}. \tag{7.3}$$

Each run contributes one eligible directed boundary, so it is selected uniformly. Only singleton runs change the boundary count. This exact drift equation identifies the information needed for a macroscopic description: the fraction of runs that are singletons, followed by the neighboring run lengths that determine the effect of a merger.

Initially, Z_0 has the law of a $\mathrm{Bin}(k, 1/2)$ variable conditioned to be even, for $k \geq 2$:

$$\mathbb{P}(Z_0 = r) = 2^{1-k} \binom{k}{r} 1\{r \text{ even}\}.$$

Each even boundary pattern corresponds to exactly two sign configurations.

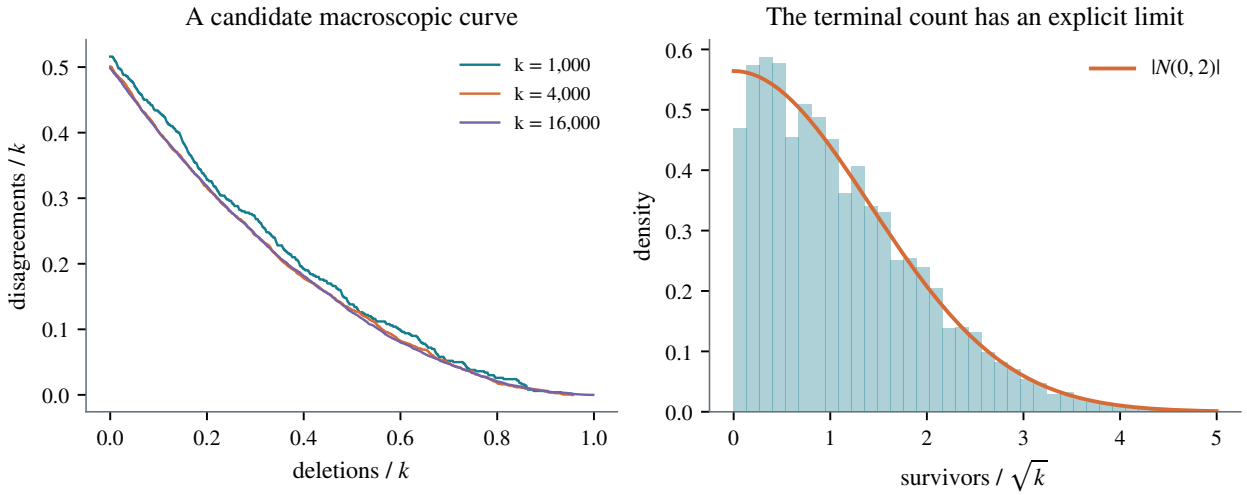


Figure 9: Left: independent simulations of the boundary count for three initial sizes. Their similarity suggests a deterministic macroscopic description. Right: 15000 terminal-count samples for $k = 2500$, obtained from the exact two-population walk, compared with the limiting half-normal density.

Question 7.4. Does $k^{-1}N_j(\lfloor kt \rfloor)$ have a deterministic limit for each j and fixed $t < 1$? Can the resulting equations describe the passage from the macroscopic boundary curve to the terminal $\sqrt{k}$ scale? A separate genealogical question assigns the mass of each deleted arrow to its successor and asks for the masses carried by the final survivors.

8 The continuous coupon collector process

A finite album is complete when every label has appeared. A circle is covered when its largest remaining hole disappears. Sorting the random centers turns this continuum question into a problem about finitely many spacings.

On a circle $\mathbb{T} = \mathbb{R}/\mathbb{Z}$ of circumference one, choose iid uniform centers $z_1, z_2, \dots$ and place closed arcs I_j of length $\theta \in (0,1)$ about them. Let $T_n = \bigcup_{j \le n} I_j$, $L_n = |T_n|$, and $\tau_\theta = \min\{n : T_n = \mathbb{T}\}$. How long does full coverage take?

Sort the centers around the circle and let $D_1, \dots, D_n$ be their circular spacings. The circle is covered exactly when $\max_i D_i \le \theta$. The spacing vector has the uniform Dirichlet law on the unit simplex: fix one sampled center as the origin and order the remaining $n-1$ uniform points. For $n \ge 2$, any specified j gaps exceed θ with probability $(1-j\theta)_+^{n-1}$. Inclusion–exclusion gives

$$\mathbb{P}(\tau_\theta \le n) = \sum_{j=0}^{n} (-1)^j \binom{n}{j} (1-j\theta)_+^{n-1}, \qquad n \ge 2. \tag{8.1}$$

For $n = 1$ the coverage probability is zero. This is the classical equal-arc coverage formula associated with Stevens; Huffer and Shepp [8] discuss its history and extensions.

Proposition 8.1. *As $\theta \downarrow 0$,*

$$\mathbb{E}\tau_\theta \sim \frac{1}{\theta} \log \frac{1}{\theta}. \tag{8.2}$$

Proof. Let $N = \lfloor 1/\theta \rfloor$ and choose N equally spaced test points. The sets of centers that cover these points are disjoint arcs of length θ, up to endpoints. Covering all test points requires collecting N types of probability θ each, with a possible null draw. Its expected time is H_N/θ, which is a lower bound for $\mathbb{E}\tau_\theta$.

For the upper bound, a union bound over the spacings gives $\mathbb{P}(\tau_\theta > n) \le n(1-\theta)^{n-1}$ for $n \ge 2$. Choose $m = \lceil (1+\varepsilon)\theta^{-1} \log(1/\theta) \rceil$. Then

$$\mathbb{E}\tau_\theta \le m + \sum_{n \ge m} n(1-\theta)^{n-1} = m + \frac{m(1-\theta)^{m-1}}{\theta} + \frac{(1-\theta)^m}{\theta^2}.$$

The last two terms are $o(\theta^{-1}\log(1/\theta))$ for fixed $\varepsilon > 0$. Let $\varepsilon \downarrow 0$. □

The same spacing formula identifies the finer transition. If $\theta_n = (\log n + x)/n$ for fixed x, then

$$\mathbb{P}(\tau_{\theta_n} \le n) \longrightarrow \exp(-e^{-x}). \tag{8.3}$$

For fixed j the jth term of inclusion–exclusion tends to $(-1)^j e^{-jx}/j!$. Finite Bonferroni bounds, followed by their length tending to infinity, justify summing these limiting terms.

The expected covered length remains simple:

$$\mathbb{E}L_n = 1 - (1-\theta)^n. \tag{8.4}$$

For two fixed points at circular distance $d \in [0, 1/2]$, the fraction of centers missing both is $q_\theta(d) = [1 - \theta - \min(d, \theta)]_+$. Thus

$$\mathbb{P}(z, w \in T_n) = 1 - 2(1-\theta)^n + q_\theta(d)^n. \tag{8.5}$$

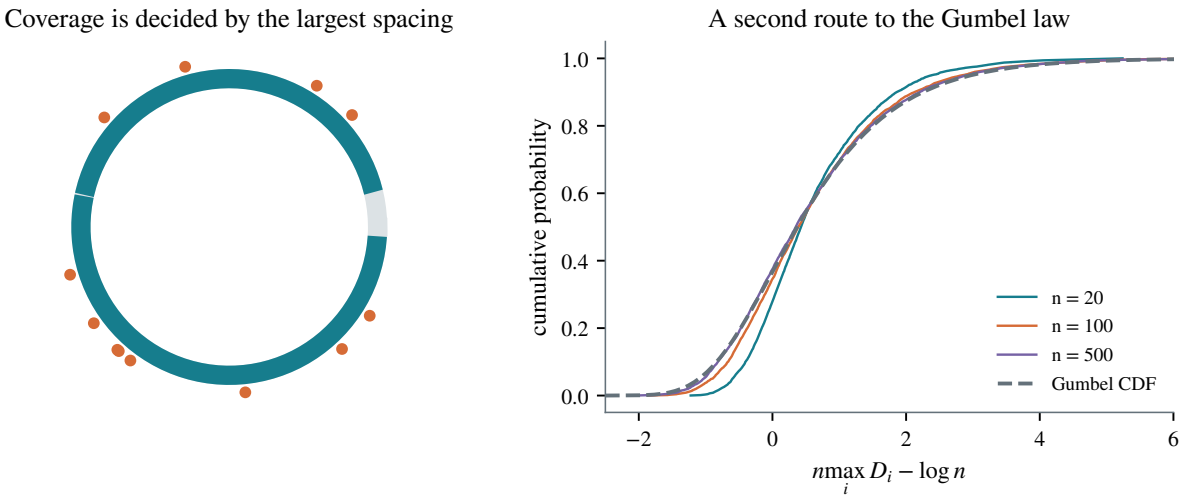


Figure 10: Left: twelve random arcs of length 0.16; orange points mark their centers and teal marks their union. Right: empirical distributions from 5000 independent circular-spacing vectors at each size, compared with the Gumbel limit for the centered largest spacing.

Integrating over their distance yields, for $n \geq 1$,

$$\mathbb{E}L_n^2 = 1 - 2(1-\theta)^n + \frac{2}{n+1}(1-\theta)^{n+1} + \frac{n-1}{n+1}(1-2\theta)_+^{n+1}, \tag{8.6}$$

$$\mathrm{Var}(L_n) = \frac{2}{n+1}(1-\theta)^{n+1} + \frac{n-1}{n+1}(1-2\theta)_+^{n+1} - (1-\theta)^{2n}. \tag{8.7}$$

As a check, $L_1 = \theta$ is deterministic. If $\theta = \alpha \log n/n$ with fixed $\alpha > 0$, (8.7) gives $\mathrm{Var}(L_n) \sim 2n^{-1-\alpha}$. Full coverage is controlled by the largest gap, which is why a small expected uncovered length alone does not settle the coverage probability.

A two-point target already displays the role of geometry. Fix two points at circular distance $d \in [0, 1/2]$, and let T_1, T_2 be the numbers of arcs needed to hit them. Each has mean $1/\theta$. A single arc hits at least one with probability $u_\theta(d) = 1 - q_\theta(d) = \min\{1, \theta + \min(d, \theta)\}$. Since the first hit has mean $1/u_\theta(d)$,

$$\mathbb{E}\max(T_1, T_2) = \frac{2}{\theta} - \frac{1}{u_\theta(d)}. \tag{8.8}$$

For $\theta \leq 1/2$, this increases from $1/\theta$ at coincidence to $3/(2\theta)$ once the separation reaches θ. Separating the targets removes the opportunity to cover both with the same arc.

Question 8.2. Suppose a set A, itself a union of length-θ arcs, has already been covered and has length l. Among such sets, which minimize or maximize the expected number of further arcs needed? Is a single arc extremal, and what role do equally spaced components play?

The first tractable case is two covered components of prescribed lengths, with their separation as the variable. Formula (8.8) supplies a basic comparison to test against the continuum optimization.

9 Asymptotics of hitting times and expected value

The graph of a nondecreasing process is a staircase. Reflecting that staircase exchanges level and time, while its vertical jumps become overshoots and its horizontal pieces become waiting intervals. This picture organizes both pathwise laws and fluctuation estimates.

Let $(X_n)_{n\geq 0}$ be nondecreasing and define $\tau_y = \inf\{n \geq 0 : X_n \geq y\}$. Pathwise,

$$\{\tau_y \leq n\} = \{X_n \geq y\}, \qquad X_{\tau_y} \geq y, \qquad \tau_{X_n} \leq n. \tag{9.1}$$

Equality $X_{\tau_y} = y$ requires that the process hit the level exactly. General positive increments can overshoot. The time τ_{X_n} records the beginning of the current plateau.

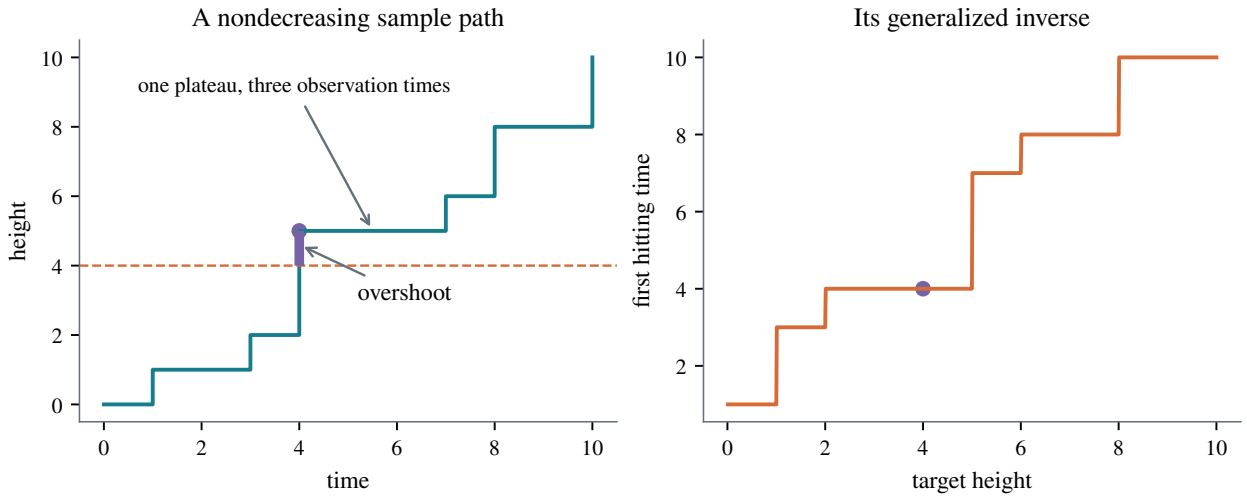


Figure 11: The same deterministic staircase viewed in two directions. The level 4 is first crossed at time 4, with overshoot 1. The plateau at height 5 has three observation times but a single first hitting time.

There are two different inverse questions: inversion of individual sample paths and inversion of their expectations. For the finite coupon collector, with N labels,

$$a(t) = \mathbb{E}X_t = N[1 - (1 - 1/N)^t], \qquad b(k) = \mathbb{E}\tau_k = N(H_N - H_{N-k}).$$

In the bulk $k/N \in [\varepsilon, 1-\varepsilon]$, these mean functions are approximately inverse after the corresponding scaling. Substituting $\mathbb{E}X_t$ into b does not compute $\mathbb{E}\tau_{X_t}$. In fact, for fixed N, τ_{X_n} eventually equals the finite collection time, so $\tau_{X_n}/n \to 0$.

For a rate-λ Poisson process, $\mathbb{E}\tau_k = k/\lambda$ for integer k, whereas the last arrival at or before time t, with $\tau_0 = 0$, has mean

$$\mathbb{E}\tau_{X_t} = \int_0^t [1 - e^{-\lambda(t-s)}]\, ds = t - \frac{1 - e^{-\lambda t}}{\lambda}. \tag{9.2}$$

This is a concrete distinction between the two inverse operations.

Proposition 9.1. *Suppose X is adapted and nondecreasing and, for some $q < 1$ and all sufficiently large n, $\mathbb{P}(X_{n+1} = X_n \mid \mathcal{F}_n) \leq q$ almost surely. Then $\tau_{X_n}/n \to 1$ almost surely.*

Proof. Fix $\varepsilon \in (0,1)$. If $\tau_{X_n} \leq \lfloor (1-\varepsilon)n \rfloor$, monotonicity forces the process to be constant between that time and n. Iterating the conditional bound gives probability at most $q^{n - \lfloor (1-\varepsilon)n \rfloor}$. These probabilities are summable. Borel–Cantelli, followed by a countable sequence $\varepsilon \downarrow 0$, proves the result. □

An unconditional one-step bound is insufficient. If B is a fair Bernoulli variable and $X_n = Bn$, then $\mathbb{P}(X_{n+1} = X_n) = 1/2$ for every n, but $\tau_{X_n} = 0$ on $\{B = 0\}$. The conditional hypothesis in Proposition 9.1 prevents this persistent dependence.

Another useful sufficient condition allows long, but relatively short, plateaus.

Proposition 9.2. *If $X_n/(cn^\alpha) \to 1$ almost surely, with constants $c > 0$, $\alpha > 0$, then*

$$\frac{\tau_y}{(y/c)^{1/\alpha}} \longrightarrow 1 \quad (y \to \infty), \qquad \frac{\tau_{X_n}}{n} \longrightarrow 1 \tag{9.3}$$

almost surely.

Proof. On any path with the stated asymptotic, eventually $(1-\varepsilon)cn^\alpha \le X_n \le (1+\varepsilon)cn^\alpha$. These inequalities bound τ_y between the corresponding inverse powers, up to integer rounding. Let $\varepsilon \downarrow 0$, then substitute $y = X_n \to \infty$. □

For sums of iid nonnegative increments with mean $\mu \in (0, \infty)$, the strong law gives $\tau_y/y \to 1/\mu$ almost surely. The elementary renewal theorem gives convergence of the means, and hence L^1 convergence as well. If the increments also have variance $\sigma^2 \in (0, \infty)$, the ordinary central limit theorem gives the inverse fluctuation scale:

$$\frac{\tau_y - y/\mu}{\sqrt{\sigma^2 y/\mu^3}} \longrightarrow_d N(0,1). \tag{9.4}$$

For $n_y = \lfloor y/\mu + z\sqrt{\sigma^2 y/\mu^3} \rfloor$, the event $\{\tau_y \le n_y\}$ is $\{S_{n_y} \ge y\}$. Its standardized threshold $(y - \mu n_y)/(\sigma\sqrt{n_y})$ tends to $-z$, proving (9.4). The factor μ^{-3} combines the variance accumulated by time y/μ with the squared reciprocal slope needed to convert height fluctuations into time fluctuations.

Processes with nonmonotone paths require information about their running maxima, since first passage is an inverse of the running maximum.

10 Number of maxima of iid random variables

For observations with atoms, a record has two attributes: its height and the number of observations sharing that height. The second attribute can retain oscillations that are almost invisible numerically. The geometric distribution gives a particularly explicit example.

Let X_i be iid, $M_n = \max_{i\le n} X_i$, and $N_n = |\{i \le n : X_i = M_n\}|$. For a continuous distribution $N_n = 1$ almost surely. With atoms there can be ties, and symmetry gives $\mathbb{P}(X_n = M_n) = \mathbb{E}N_n/n$. For an integer-valued law, write $p_k = \mathbb{P}(X_1 = k)$, $F_k = \mathbb{P}(X_1 \le k)$. Then, for $1 \le m \le n$,

$$\mathbb{P}(N_n = m) = \sum_k \binom{n}{m} p_k^m F_{k-1}^{n-m}, \tag{10.1}$$

$$\mathbb{E}N_n = n \sum_k p_k F_k^{n-1}, \tag{10.2}$$

$$\mathbb{P}(X_n > M_{n-1}) = \sum_k p_k F_{k-1}^{n-1}. \tag{10.3}$$

Summing the first identity over m telescopes to $\sum_k (F_k^n - F_{k-1}^n) = 1$. More generally,

$$\mathbb{E}(N_n)_r = (n)_r \sum_k p_k^r F_k^{n-r}, \quad 1 \le r \le n. \tag{10.4}$$

For geometric probabilities $p_k = pq^k$, $k \ge 0$, $q = 1-p$, the expectation becomes $n\sum_{k\ge0} pq^k(1 - q^{k+1})^{n-1}$. Its apparent limiting constant hides a small oscillation.

Proposition 10.1. *Let $a = 1/q$. Along a sequence with fractional parts $\{\log_a n\} \to r \in [0,1]$,*

$$\mathbb{E}N_n \longrightarrow H(r) := \frac{p}{q} \sum_{j\in\mathbb{Z}} a^{r-j} e^{-a^{r-j}}. \tag{10.5}$$

The function H is continuous, one-periodic, and nonconstant. Its average over one period is $p/[q\log(1/q)]$.

Proof. Put $m = \lfloor \log_a n \rfloor$ and replace $k+1$ by $m+j$ in (10.2). Each summand becomes $(p/q)x(1 - x/n)^{n-1}$ with $x = a^{\{\log_a n\}-j}$. For $n \ge 2$, it is bounded by $(p/q)xe^{-x/2}$, and the small-x tail is bounded by $(p/q)x$. These bounds are summable uniformly over the phase $\{\log_a n\} \in [0,1]$, so dominated convergence gives the limit. Periodicity is a shift of j. Substituting $x = a^{r-j}$ over a complete period gives the average $(p/q)(\log a)^{-1}\int_0^\infty e^{-x}\,dx$. The Fourier coefficient at nonzero integer h is

$$\frac{p}{q\log a}\int_0^\infty e^{-x}x^{-2\pi ih/\log a}\,dx = \frac{p}{q\log a}\Gamma\left(1 - \frac{2\pi ih}{\log a}\right),$$

which is nonzero. Thus H is not constant. □

For $p = q = 1/2$, the average is $1/\log 2 \approx 1.442695$. Figure 12 resolves the small periodic departure from that average. The persistent phase is the fractional part of $\log_2 n$. Lattice oscillations for maxima and their ties form part of the discrete extreme-value theory discussed by Ottolini [9].

The strict-record probability has its own tail-sensitive behavior. Even with full support, $n\mathbb{P}(X_n > M_{n-1})$ can have subsequences tending to zero. Suppose along levels $j \to \infty$ that $r_j := \mathbb{P}(X_1 > j) = o(p_j)$, and take $n_j = \lfloor (p_j r_j)^{-1/2} \rfloor$. Then $n_j p_j \to \infty$ and $n_j r_j \to 0$. The probability of a unique maximum is at most

$$n_j r_j + n_j p_j (1-p_j)^{n_j - 1} + (1-p_j)^{n_j} \longrightarrow 0. \tag{10.6}$$

These three terms cover a maximum above j, a unique value at j, or all observations below j. By symmetry, this probability is $n_j\mathbb{P}(X_{n_j} > M_{n_j-1})$. A Poisson law with fixed positive mean satisfies $r_j/p_j \to 0$, so it already provides such a counterexample.

There is a sharper conclusion for rapidly thinning masses.

Proposition 10.2. *Suppose $p_j > 0$ eventually and $p_{j+1}/p_j \to 0$. If $R_n = n\mathbb{P}(X_n > M_{n-1})$, then*

$$\liminf_n R_n = 0, \qquad \limsup_n R_n = e^{-1}.$$

Every value in $[0, e^{-1}]$ is a subsequential limit. In particular this holds for every fixed positive Poisson mean.

Proof. Write $q_j = \mathbb{P}(X_1 > j)$. The ratio assumption gives $q_j \sim p_{j+1}$. For fixed $c > 0$, take $n_j = \lfloor c/p_j \rfloor$. The probability of a unique maximum at j tends to ce^{-c}. A maximum above j has probability at most $n_j q_j = o(1)$. A unique maximum below j requires at most one observation at or above $j-1$, whose expected count tends to infinity; the corresponding binomial probability tends to zero. Thus $R_{n_j} \to ce^{-c}$. For a zero subsequence take instead $n_j = \lfloor (p_j p_{j+1})^{-1/2} \rfloor$. Then $n_j q_j \to 0$ and $n_j p_j \to \infty$: a maximum above j is unlikely, while a unique maximum at or below j requires at most one observation at or above j, also unlikely. Since ce^{-c} ranges over $(0, e^{-1}]$, all asserted subsequential limits follow.

For the upper bound choose $j = j(n)$ with $nq_j \le 1 < nq_{j-1}$. A maximum above $j+1$ has probability $o(1)$, because $q_{j+1}/q_j \to 0$. A unique maximum below j again requires at most one observation at or above $j-1$, and $nq_{j-2} \to \infty$. Only the levels $j, j+1$ therefore contribute, up to $o(1)$. Put $u = np_j$, $v = np_{j+1}$, so $v/u \to 0$ and $v \le 1$. Along a subsequence where u is bounded, the upper-level contribution vanishes and the lower is at most $e^{-1} + o(1)$, using $u(1-q_{j-1})^{n-1} \le ue^{-(1-1/n)u}$. Along a subsequence where $u \to \infty$, the lower contribution vanishes and the upper is bounded in the same way by $e^{-1} + o(1)$. Every sequence has a subsequence of one of these two types, proving the upper bound. □

This two-level calculation makes the record consequence of discrete extreme-value oscillations explicit; compare Ottolini's treatment of maxima and ties [9].

For a discrete uniform law on $\{1, \dots, k\}$,

$$\mathbb{P}(X_n > M_{n-1}) = \frac{1}{k}\sum_{i=0}^{k-1}(i/k)^{n-1} \sim \frac{1}{k}(1-1/k)^{n-1} \tag{10.7}$$

when $k \ge 2$ is fixed and $n \to \infty$. If $k = k_n$ and $k_n/n \to \alpha > 0$, then instead

$$n\mathbb{P}(X_n > M_{n-1}) \longrightarrow \frac{1}{\alpha(e^{1/\alpha} - 1)}. \tag{10.8}$$

To see this, index the sum by distance r below the top: each term tends to $\alpha^{-1}e^{-r/\alpha}$, $r \ge 1$, and a geometric bound justifies summing.

10.1 Record process

The pair (M_n, N_n) is Markov. Given $(M_n, N_n) = (m, r)$, the count becomes 1 with probability $\mathbb{P}(X_1 > m)$, stays r with probability F_{m-1}, and becomes $r+1$ with probability p_m. In the first case the new maximum has the law of X_1 conditional on exceeding m.

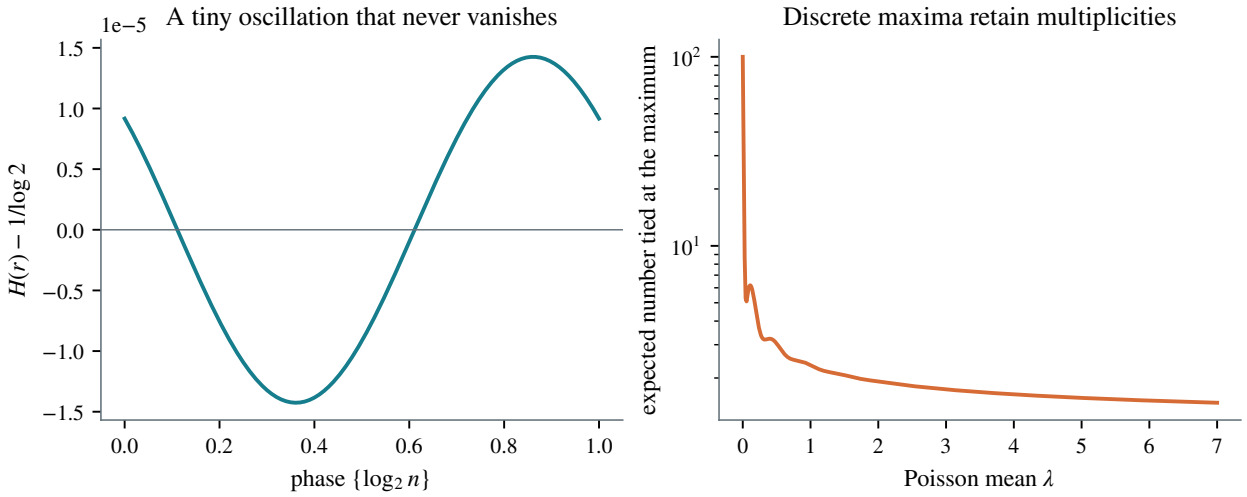


Figure 12: Left: the periodic geometric multiplicity limit for $p = 1/2$, after subtracting its phase average; the vertical scale resolves the small oscillation. Right: the expected number tied at the maximum of 100 Poisson observations as their common mean varies, computed from (10.2).

For the geometric law, the height and multiplicity have a joint limit along a fixed phase. Put $m_n = \lfloor \log_a n \rfloor$ and suppose $\{\log_a n\} \to r \in [0, 1)$. The numbers of observations at levels $m_n + j$ converge, on every finite set of indices, to independent Poisson variables with means pa^{r-j}. Thus, for $j \in \mathbb{Z}$ and $b \geq 1$,

$$\mathbb{P}(M_n - m_n = j, N_n = b) \longrightarrow e^{-a^{r-j}} \frac{(pa^{r-j})^b}{b!}. \tag{10.9}$$

The exponential factor requires zero observations above j as well as the Poisson mass at j: the sum of the limiting means at levels $j, j+1, \ldots$ is a^{r-j}. The upper tail of the maximum is bounded by the expected number above a high level; the lower tail has limiting probability $\exp(-a^{r-j-1})$. These two bounds make the finite-window Poisson argument tight and establish the joint limit. Summing (10.9) gives both the maximum law and the multiplicity law. Keeping the phase in the state space turns an oscillating sequence into an explicit family of limits.

Question 10.3. Which tail assumptions give a comparable joint description for the entire record process and the counts at nearby levels? How does dependence between observations alter the Poisson level counts in (10.9)?

The next section develops the continuous-height counterpart, where a Poisson cloud yields the whole trajectory of the running maximum.

11 Functional limit for the extreme-value process

Watch a growing sample while continually recentering its maximum. Between records the centered value drifts down; a new record sends it upward. For exponential observations, logarithmic time makes this sawtooth process stationary. Its transition law, simulation rule, and covariance can all be written explicitly.

More generally, if $b_n(M_n - a_n) \to_d L$ for iid observations, with $b_n > 0$, the process question concerns $b_{\lfloor nt \rfloor}(M_{\lfloor nt \rfloor} - a_{\lfloor nt \rfloor})$, $t > 0$. We develop the exponential example from its Poisson construction.

11.1 Exponential observations

Take $Y_i \sim \mathrm{Exp}(1)$ and let Π be a Poisson point process on $(0,\infty) \times \mathbb{R}$ with intensity $dt\, e^{-x} dx$. Define

$$E(t) = \sup\{x : (s,x) \in \Pi,\ 0 < s \le t\}, \qquad \xi(t) = E(t) - \log t. \tag{11.1}$$

Then $\mathbb{P}(E(t) \le x) = \exp(-te^{-x})$, so every $\xi(t)$ has the standard Gumbel law $G(x) = \exp(-e^{-x})$.

The sample point process $\sum_i \delta_{(i/n, Y_i - \log n)}$ converges to Π on bounded time intervals above any fixed height. Indeed, for a time interval of length h, the exceedance probability above height x is e^{-x}/n per observation, and the independent rare-event counts converge to Poisson counts with mean he^{-x}; the same argument for disjoint rectangles gives their joint limit. There are finitely many points above each height on a bounded interval. On $[\varepsilon, T]$, choose that height low enough that the maximum at ε exceeds it with arbitrarily high probability. The running-maximum map then depends on finitely many points, whose times and heights have no ties. Matching those points proves convergence in the Skorokhod J_1 topology. Consequently

$$M_{\lfloor nt \rfloor} - \log \lfloor nt \rfloor \ \longrightarrow_d\ \xi(t) \quad \text{on every } [\varepsilon, T] \subset (0,\infty). \tag{11.2}$$

For $0 < s < t$, the maximum of the new Poisson points is independent of the past. The transition law is therefore

$$\mathbb{P}(\xi(t) \le y \mid \xi(s) = x) = \begin{cases} 0, & y < x - \log(t/s), \\ \exp(-(1 - s/t)e^{-y}), & y \ge x - \log(t/s). \end{cases} \tag{11.3}$$

At the lower endpoint there is an atom of mass $\exp(-(t/s - 1)e^{-x})$, corresponding to no new record.

Put $U(w) = \xi(e^w)$. The scaling $(t,x) \mapsto (ct, x + \log c)$ preserves the Poisson intensity, so U is stationary. For $h > 0$,

$$\mathbb{P}(U(w+h) \le y \mid U(w) = u) = \begin{cases} 0, & y < u - h, \\ \exp(-(1 - e^{-h})e^{-y}), & y \ge u - h. \end{cases} \tag{11.4}$$

It moves down at speed one between positive jumps. At state u, its jump rate is e^{-u} and its jump height is an independent $\mathrm{Exp}(1)$ variable. For a smooth bounded test function, the generator is

$$\mathcal{L}f(u) = -f'(u) + e^{-u} \int_0^\infty [f(u+j) - f(u)] e^{-j}\, dj. \tag{11.5}$$

This gives an exact simulation using ordinary-sized numbers. At current state u, draw independent $E, J \sim \mathrm{Exp}(1)$ and set

$$H = \log(1 + e^u E), \qquad U(w+H) = u - H + J. \tag{11.6}$$

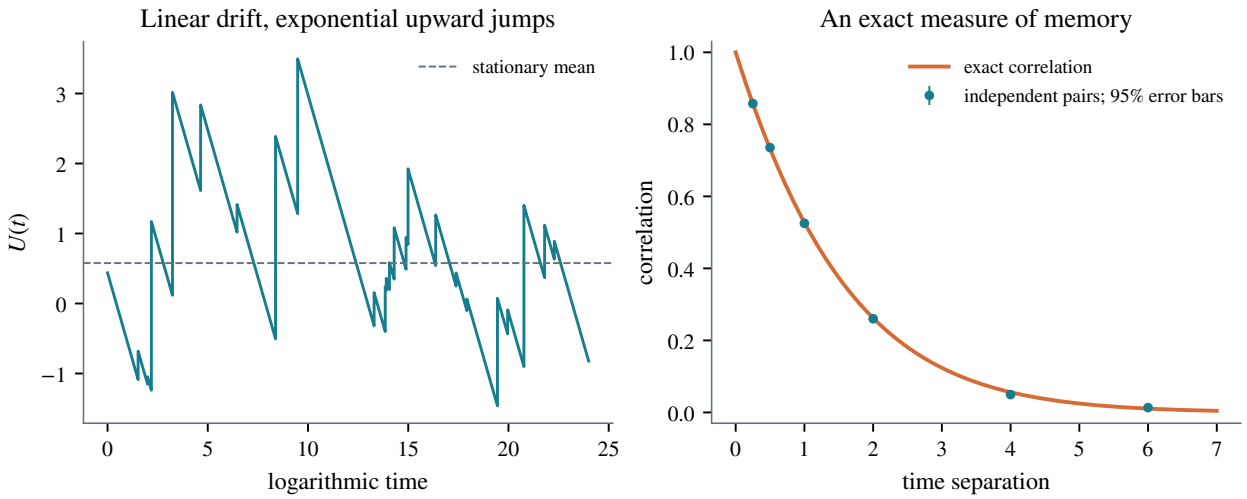


Figure 13: Left: a stationary path generated by the exact event rule (11.6). Right: the correlation obtained from (11.8), compared with 150000 independent coupled pairs at each displayed lag. Error bars are 1.96 estimated standard errors.

During the wait, $U(w+s) = u - s$ for $0 \le s < H$. The integrated jump rate is $e^{-u}(e^s - 1)$, which explains the formula for H. Numerically one can evaluate it as $\mathrm{logaddexp}(0, u + \log E)$. Starting with $U(0) = -\log E_0$ for an independent $E_0 \sim \mathrm{Exp}(1)$ gives the stationary law.

At a fixed current state, a larger upward jump reduces the rate of the next jump. Unconditionally, the jump times in logarithmic time form a rate-one Poisson process. One can see this from the original iid sample: its record indicators are independent Bernoulli variables with probabilities $1/i$, because successive relative ranks are independent and uniform on $[i]$. On logarithmic intervals the sums of these probabilities tend to interval lengths and their maxima tend to zero, giving independent Poisson counts. The dependence between jump times and observed values remains visible in (11.4).

There is a particularly simple coupling of two stationary observations. Let V, E be independent rate-one exponentials and put $r = e^{-h}$. Then

$$U(0) = -\log V, \qquad U(h) = -\log \min\left\{\frac{V}{r}, \frac{E}{1-r}\right\}. \tag{11.7}$$

The minimum is again rate-one exponential. This both verifies stationarity and quantifies the memory retained from V.

Proposition 11.1. *For the stationary process and $h \ge 0$,*

$$C(h) := \mathrm{Cov}(U(0), U(h)) = \int_h^\infty \frac{s}{e^s - 1}\, ds. \tag{11.8}$$

In particular, $C(0) = \pi^2/6$ and $C(h) \sim (h+1)e^{-h}$ as $h \to \infty$.

Proof. Write $\gamma = \mathbb{E}U(0)$. The conditional infinitesimal mean increment at state u is $-1 + e^{-u}$. Consequently, for $h > 0$,

$$C'(h) = \mathbb{E}[(U(0) - \gamma)e^{-U(h)}].$$

This identity can also be obtained by differentiating the conditional mean in (11.7); on compact subintervals of $h > 0$ the integrands are dominated by integrable functions of V and $|\log V|$. Conditionally on $V = v$,

$$\mathbb{E}[e^{-U(h)} \mid V = v] = \frac{1 - e^{-(1-r)v/r}}{1-r}.$$

The integral $\int_0^\infty (-\log v) e^{-v/r}\,dv = r(\gamma - \log r)$ now gives

$$C'(h) = \frac{r\log r}{1-r} = -\frac{h}{e^h - 1}.$$

As $h \to \infty$, the minimum in (11.7) tends almost surely to E. The fixed Gumbel marginals have finite fourth moments, so products are uniformly integrable and $C(h) \to 0$. Integrating the derivative proves the formula for $h > 0$, and continuity gives the value at zero. Finally, expand $(e^s - 1)^{-1} = \sum_{j\ge 1} e^{-js}$; the first summand gives $(h+1)e^{-h}$ and the rest are of smaller order. □

The covariance also controls accumulated fluctuations. For $A_T = \int_0^T (U(t) - \gamma)\,dt$, stationarity gives

$$\operatorname{Var}(A_T) = 2\int_0^T (T-h)C(h)\,dh = 4\zeta(3)T + O(1). \tag{11.9}$$

Indeed, Tonelli's theorem gives $\int_0^\infty C(h)\,dh = \int_0^\infty s^2/(e^s-1)\,ds = 2\zeta(3)$, and $\int_0^\infty hC(h)\,dh < \infty$.

Theorem 11.2. *For stationary U, as $T \to \infty$,*

$$\left(\frac{1}{\sqrt{T}}\int_0^{Tt} (U(s) - \gamma)\,ds\right)_{0\le t\le 1} \longrightarrow_d \left(\sqrt{4\zeta(3)}\,B(t)\right)_{0\le t\le 1}$$

in $C[0,1]$, where B is standard Brownian motion.

Proof. Construct two stationary versions with independent initial exponential variables V, V', sharing all future Poisson points. At time h the old minimum wins against the new points with probability e^{-h} for either version, by (11.7). Otherwise both versions have the same state and remain identical thereafter. The second future is independent of the first past. The coupling bound therefore gives strong mixing coefficient $\alpha(h) \le 2e^{-h}$.

Set $Y_j = \int_j^{j+1} (U(s) - \gamma)\,ds$. This is a stationary centered sequence with $\mathbb{E}|Y_0|^4 \le \mathbb{E}|U(0) - \gamma|^4 < \infty$ by Jensen's inequality, and $\alpha_Y(k) \le 2e^{-(k-1)}$ for $k \ge 2$. The strong-mixing invariance principle applies. More explicitly, the criterion in Doukhan–Massart–Rio [10, Theorem 1] is $\int_0^1 \alpha_Y^{-1}(u) Q_Y(u)^2\,du < \infty$, where Q_Y is the upper-tail quantile of $|Y_0|$. Here $\alpha_Y^{-1}(u) = O(1 + |\log u|)$ and $Q_Y(u) \le (\mathbb{E}|Y_0|^4/u)^{1/4}$, so the integral is finite. The limiting variance is $\lim_n n^{-1}\operatorname{Var}(A_n) = 4\zeta(3)$ by (11.9).

Finally, let $Z_j = \int_j^{j+1} |U(s) - \gamma|\,ds$. Stationarity and the fourth-moment bound give

$$\mathbb{P}\left(\max_{0\le j\le n} Z_j > \varepsilon\sqrt{n}\right) \le \frac{(n+1)\mathbb{E}Z_0^4}{\varepsilon^4 n^2} \longrightarrow 0.$$

This controls the partial last blocks uniformly in time. It transfers the invariance principle from block sums to the continuous integral; the limit is continuous, giving convergence in the uniform topology. □

Question 11.3. From a prescribed starting state, what is the distribution of the first passage above a high level? Can the same coupling give useful finite-time error bounds for the integral's Gaussian approximation?

For an exact Pareto law $\mathbb{P}(Y_i > y) = y^{-\alpha}$, $y \ge 1$, the variables $\alpha \log Y_i$ are exponential. Hence the corresponding time-normalized maximum has limit

$$\frac{M_{\lfloor nt\rfloor}}{(nt)^{1/\alpha}} \longrightarrow_d \exp(U(\log t)/\alpha).$$

12 Ruler distribution

Sampling from a density favors locations where the density is large. Now record the density's height at the sampled location. This produces a new distribution, whose moments are precisely the power integrals of the starting density.

Let f be a probability density on $\mathbb{R}$, let Y have that density, and define $X_f = f(Y)$. Its distribution is

$$\mathbb{P}(X_f \le x) = \int_{\mathbb{R}} f(t) 1\{f(t) \le x\}\, dt. \tag{12.1}$$

One can picture a ruler at height x, retaining the parts of the graph whose height is at most x and integrating under those parts. The random variable is the density evaluated at a point sampled from that same density.

For any measurable nonnegative φ, the change-of-variable definition immediately gives

$$\mathbb{E}\varphi(X_f) = \int_{\mathbb{R}} \varphi(f(t)) f(t)\, dt. \tag{12.2}$$

In particular, all power moments have the same simple form:

$$\mathbb{E}X_f^\alpha = \int_{\{f>0\}} f(t)^{\alpha+1}\, dt, \qquad \alpha \in \mathbb{R}, \tag{12.3}$$

with either side allowed to be infinite. Integrating only over $\{f > 0\}$ specifies the convention when negative powers are used. For $\alpha = 1$, $\mathbb{E}X_f = \int f^2$.

The density of X_f, when it exists, involves the slopes of f. If f is C^1 and has finitely many strictly monotone pieces, then at regular values $x > 0$ the absolutely continuous part has density

$$g_f(x) = x \sum_{t: f(t)=x} \frac{1}{|f'(t)|}. \tag{12.4}$$

This is the usual one-dimensional change-of-variables formula on each monotone piece. It gives the whole law if the set where $f' = 0$ has zero probability under $f(t)dt$. A plateau of positive length a at height $c > 0$ instead creates an atom of mass ca at c. More generally, a critical set can contribute a singular part; counting inverse images alone is not a density formula.

For example, a uniform density on an interval of length a gives $X_f = 1/a$ almost surely. For $f(t) = \lambda e^{-\lambda t} 1\{t \ge 0\}$, $e^{-\lambda Y}$ is uniform on $(0, 1)$, so X_f is uniform on $(0, \lambda)$.

For the triangular density $f(y) = (1 - |y|)1\{|y| \le 1\}$, the two inverse branches at height $x \in (0, 1)$ each have slope of magnitude one. Thus $g_f(x) = 2x$, and $\mathbb{P}(X_f \le x) = x^2$ on $[0, 1]$. The probability of a sampled height below x equals the area of the two small outer triangles.

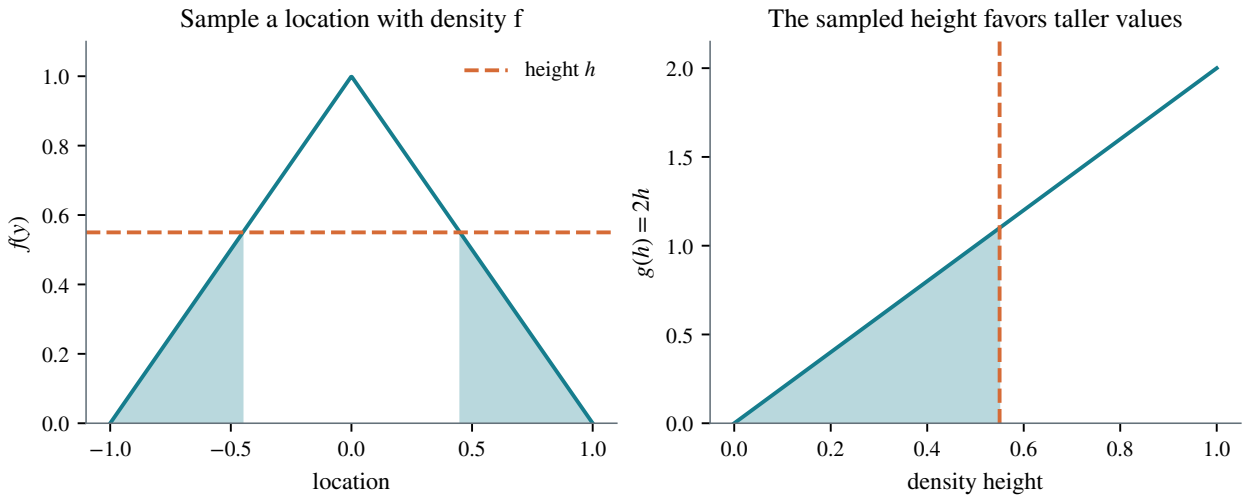


Figure 14: The triangular density and its sampled-height distribution. Both shaded areas equal h^2 with $h = 0.55$. The change of variables weights each height by the probability mass at its inverse images.

Translation of a density leaves the sampled-height law unchanged. Scaling space by $a > 0$ gives $f_a(y) = a^{-1} f(y/a)$ and $X_{f_a} \overset{d}{=} a^{-1} X_f$. Thus the construction distinguishes the shape of a density from its choice of spatial units.

The discrete version uses counting measure in place of Lebesgue measure. If Y has masses (p_j), define $X = p_Y$. Then

$$\mathbb{P}(X = c) = \sum_{j : p_j = c} p_j, \qquad \mathbb{E}X^{\alpha} = \sum_{j : p_j > 0} p_j^{\alpha+1}. \tag{12.5}$$

A density and a collection of point masses are different choices of base measure; this formulation covers both without treating Dirac masses as function values. The power integrals in (12.3) also connect the construction to Rényi entropy, while $-\mathbb{E} \log X_f = -\int f \log f$ is differential entropy whenever that integral is defined.

13 Random decreasing sequences

Repeated multiplication produces a decreasing random sequence. Summing that sequence leads to a fixed-point equation; taking logarithms instead leads to arrival times. For uniform factors, these two descriptions meet in the Dickman distribution.

Let X_i be iid in $[0,1]$, with $\mathbb{P}(X_1 = 1) < 1$, and define $Y_n = \prod_{i=1}^n X_i$, $S = \sum_{n\geq 1} Y_n$. Writing $\mu = \mathbb{E}X_1 < 1$,

$$\mathbb{E}Y_n^r = (\mathbb{E}X_1^r)^n \quad (r > 0), \qquad \mathbb{E}S = \frac{\mu}{1-\mu} < \infty. \tag{13.1}$$

Thus S is finite almost surely. To obtain probabilities $p_n = Y_n/S$ that are all positive, additionally assume $X_1 > 0$ almost surely. If zero is allowed, $S = 0$ whenever $X_1 = 0$, and normalization requires a separate convention.

If X_1 has density f and g_n denotes the density of Y_n, then

$$g_1 = f, \qquad g_n(y) = \int_y^1 f(y/x) g_{n-1}(x) \frac{dx}{x}. \tag{13.2}$$

Taking logarithms turns products into sums. Under $\mathbb{E}|\log X_1|^2 < \infty$, the central limit theorem approximates $\log Y_n$ by a normal variable with mean $n\mathbb{E}\log X_1$ and variance $n\,\mathrm{Var}(\log X_1)$. The products for different n share their early factors, so separate lognormal marginals do not specify a distribution for their sum.

The sum has a useful exact description as a perpetuity:

$$S \stackrel{d}{=} X(1 + S'), \tag{13.3}$$

where S' is an independent copy of S, independent of X. Both terms on the right use the same factor X. If F is the CDF of X and H that of S, then

$$H(s) = \int_{[0,\infty)} F\left(\frac{s}{1+t}\right) dH(t), \qquad s \geq 0, \tag{13.4}$$

where $F(u) = 1$ for $u \geq 1$.

The fixed-point equation also gives exact moments. With $\nu = \mathbb{E}X^2$,

$$\mathbb{E}S^2 = \frac{\nu(1+\mu)}{(1-\mu)(1-\nu)}, \qquad \mathrm{Var}(S) = \frac{\nu - \mu^2}{(1-\mu)^2(1-\nu)}. \tag{13.5}$$

More generally, setting $m_r = \mathbb{E}S^r$ and $m_0 = 1$,

$$m_r = \frac{\mathbb{E}X^r}{1 - \mathbb{E}X^r} \sum_{j=0}^{r-1} \binom{r}{j} m_j, \qquad r = 1, 2, \ldots. \tag{13.6}$$

Finiteness follows first for the partial sums: their corresponding recursion and induction give uniform bounds; monotone convergence then gives the identity for S.

For uniform X_i, the product density is particularly simple:

$$g_n(y) = \frac{(-\log y)^{n-1}}{(n-1)!}, \qquad 0 < y < 1.$$

The sum S has the Dickman distribution, a well-studied special case of a Vervaat perpetuity [11]. Here $\mathbb{E}S = 1$ and $\operatorname{Var} S = 1/2$. Indeed, $-\log X_i$ are exponential waiting times, so $Y_n = e^{-T_n}$ at the arrival times T_n of a rate-one Poisson process. Its Laplace functional gives

$$\mathbb{E}e^{-tS} = \exp\left(\int_0^1 \frac{e^{-tu}-1}{u}\,du\right), \qquad t \geq 0. \tag{13.7}$$

The density h is constant on $(0,1]$ and satisfies $xh'(x) = -h(x-1)$ for $x > 1$. To see the equation directly, (13.3) says that, conditional on $S' = s$, the new sum is uniform on $(0, 1+s)$. Hence $h(x) = \int_{(x-1)_+}^{\infty} h(s)/(1+s)\,ds$, and differentiation gives the claim. The normalizing constant is $h(x) = e^{-\gamma}$ on $(0,1]$ [11].

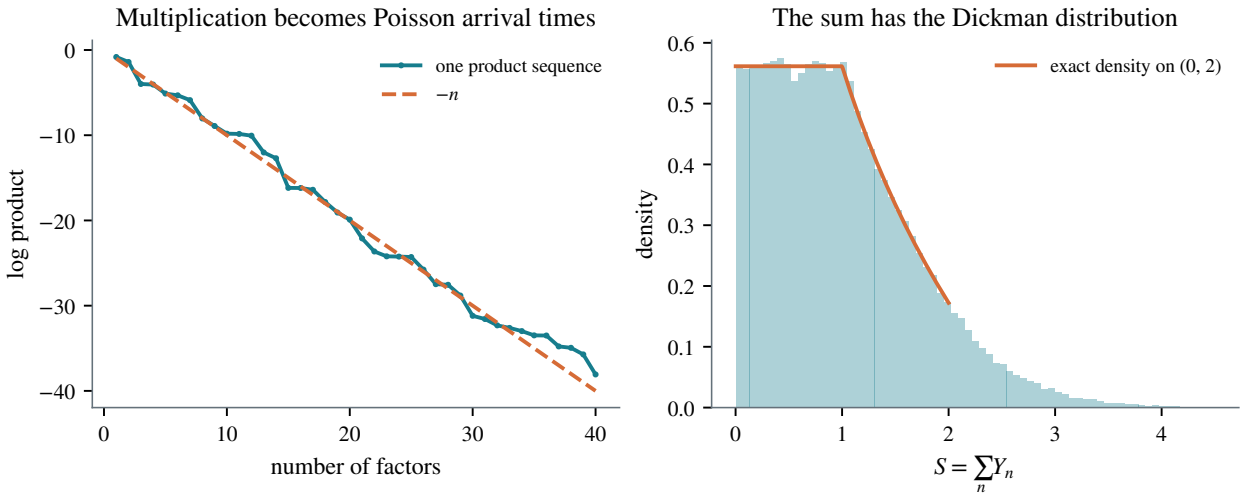


Figure 15: Uniform factors. Left: the logarithm of one product sequence, compared with its mean $-n$. Right: 90000 samples of the first 60 terms of the sum; the expected omitted tail is 2^{-60}. The orange curve is the exact Dickman density on $(0,2)$: $e^{-\gamma}$ up to one, then $e^{-\gamma}(1 - \log x)$.

The same construction gives a full one-parameter family. If X has density $\alpha x^{\alpha-1}$ on $(0,1)$, then $-\log X$ is exponential of rate α. The points T_n now form a rate-α Poisson process, so

$$\mathbb{E}e^{-tS} = \exp\left(\alpha\int_0^1 \frac{e^{-tu}-1}{u}\,du\right), \qquad \mathbb{E}S = \alpha, \quad \operatorname{Var}(S) = \frac{\alpha}{2}.$$

The parameter changes the arrival intensity while retaining the same exponential weighting of arrival times. This is the generalized Dickman family; the fixed-point viewpoint is treated in [11].

Question 13.1. For other factor laws F, how can the fixed-point equation be used to sample S efficiently and describe the normalized sequence (Y_n/S)? How does the tail of S depend on the mass of F near one?

14 Fisherman's dilemma

An unsuccessful cast supplies information: the waiting time has already exceeded the time spent. Depending on the waiting-time law, that can make the present cast more promising or less promising than a fresh one. The hazard rate describes this comparison, while a renewal equation gives its cost in total fishing time.

Each cast starts an independent waiting time $X > 0$ with CDF F and survival function $\overline{F}(t) = \mathbb{P}(X > t)$. We first assume that this law is known and minimize the expected time until the first catch.

Fix a timeout $t > 0$ with $F(t) > 0$: recast after time t if no bite has occurred, and count a bite at exactly t as a success. One attempt has duration $X \wedge t$ and success probability $F(t)$. Summing over independent failed attempts gives

$$\mu(t) := \mathbb{E}T = \frac{\mathbb{E}(X \wedge t)}{F(t)} = \frac{\int_0^t \overline{F}(u)\,du}{1 - \overline{F}(t)}. \tag{14.1}$$

This includes $t = \infty$ when the relevant expectation is finite. Under repeated use of a fixed timeout, the renewal theorem gives a long-run catch rate of $1/\mu(t)$, so minimizing the mean also maximizes that rate.

Constant timeouts suffice for this objective, up to approximation when the infimum is not attained. To make the statement precise, let $m_* = \inf_t \mu(t)$. At the start of an attempt, a timeout may depend on previous attempts and fresh randomization, but not on that cast's unknown waiting time. Conditional on that history and timeout, the expected attempt duration is at least m_* times its success probability. Sum over the attempts that are reached. For a strategy that eventually succeeds almost surely, Tonelli's theorem gives $\mathbb{E}T \geq m_* \sum_i \mathbb{P}(\text{success on attempt } i) = m_*$. Fixed timeouts with $\mu(t) \downarrow m_*$ approach this bound. The only observation within a failed cast is that the bite has not yet occurred, so a rule based on elapsed waiting alone is covered by such a timeout.

If F has density f and hazard $h(t) = f(t)/\overline{F}(t)$, then wherever $0 < F(t) < 1$,

$$\mu'(t) = \frac{\overline{F}(t)}{F(t)}[1 - h(t)\mu(t)]. \tag{14.2}$$

An interior optimum therefore satisfies $h(t)\mu(t) = 1$. Endpoints, atoms, and points where the derivative is undefined must also be considered. If $h(t) \to h(0+) \in (0, \infty)$ at zero, then $\mu(t) \to 1/h(0+)$ as $t \downarrow 0$.

There is a useful interpretation of the derivative. Since $F(t) = \int_0^t h(u)\overline{F}(u)\,du$, the reciprocal $1/\mu(t)$ is the $\overline{F}$-weighted average hazard over $[0, t]$. An increasing hazard therefore gives $\mu'(t) \leq 0$; waiting longer helps. A decreasing hazard gives $\mu'(t) \geq 0$; shorter timeouts help when recasting is free.

For an exponential waiting time of rate λ, every timeout gives $\mu(t) = 1/\lambda$. Conversely, if $\mu(t)$ is this constant for all $t > 0$, then (14.1) implies $F(t) = \lambda \int_0^t \overline{F}(u)\,du$. Thus F is absolutely continuous and $\overline{F}' = -\lambda\overline{F}$, with $\overline{F}(0) = 1$, forcing the exponential law.

A fixed setup cost $c > 0$ for *each* attempt, including the successful one, changes the mean to

$$\mu_c(t) = \frac{c + \int_0^t \overline{F}(u)\,du}{F(t)}. \tag{14.3}$$

The same argument proves optimality among history-dependent timeouts, and the interior condition becomes $h(t)\mu_c(t) = 1$. For exponential X, $\mu_c(t) = 1/\lambda + c/(1 - e^{-\lambda t})$, so waiting indefinitely is

best. If setup is charged only after a failed attempt, the numerator instead contains $c\overline{F}(t)$; the cost convention is part of the model.

For a concrete example, suppose a cast has an exponential waiting time of rate 2 with probability 0.7 and of rate 0.1 with probability 0.3. Then

$$\overline{F}(t) = 0.7e^{-2t} + 0.3e^{-0.1t}.$$

Long unsuccessful waits make the slow component increasingly likely, so the hazard decreases. Free recasting favors very short timeouts. A positive setup cost makes such rapid recasting expensive and can select a finite interior optimum, as Figure 16 illustrates.

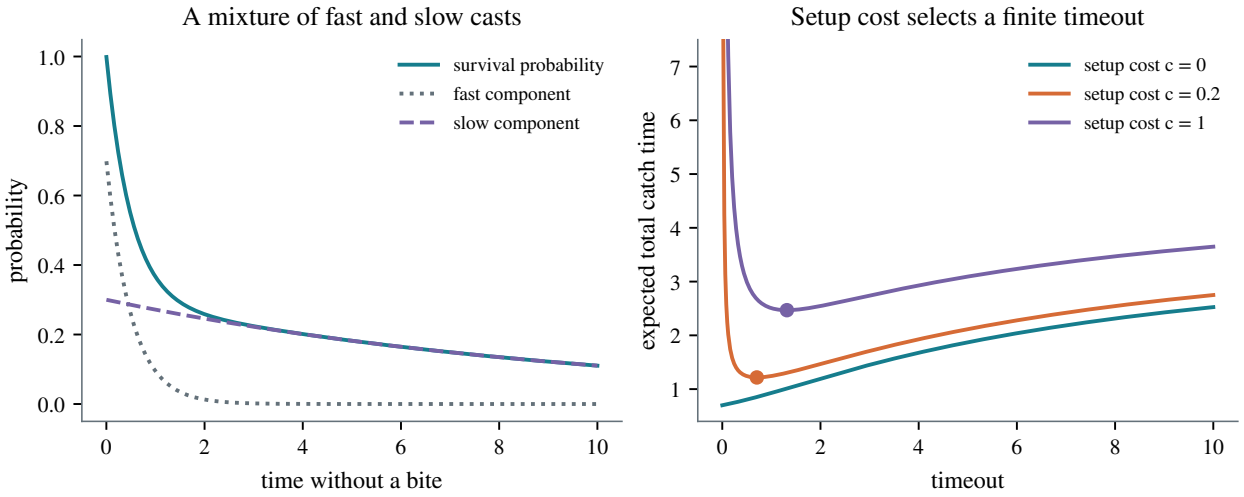


Figure 16: A mixture of fast and slow exponential waits. Left: the survival function and its two weighted components. Right: expected catch time under a fixed timeout, for three setup costs paid on every attempt. Dots mark numerical minima of the explicit formula.

The entire law of the catch time is available through its Laplace transform. For the zero-cost model and $s \geq 0$,

$$\mathbb{E}e^{-sT} = \frac{\int_{(0,t]} e^{-su}\, dF(u)}{1 - \overline{F}(t)e^{-st}}. \tag{14.4}$$

The denominator sums the geometric series of failed casts, and the numerator accounts for the successful waiting time. Equivalently, for $z > 0$ with $\overline{F}(t)z^t < 1$,

$$\mathbb{E}z^T = 1 + \frac{\log z}{1 - \overline{F}(t)z^t} \int_0^t z^u \overline{F}(u)\, du. \tag{14.5}$$

The transform equals one at $z = 1$. Derivatives of the Laplace transform at $s = 0$ give the moments whenever the relevant integrability holds.

Restarting a randomized algorithm leads to the same optimization: a fixed timeout repeats one running-time budget. When the law is unknown, Luby, Sinclair, and Zuckerman [12] construct a universal restart schedule with logarithmic overhead in the discrete running-time setting.

Question 14.1. If the waiting-time law is unknown, how should observations of bites and censored waits be used to learn a good timeout? How does a finite fishing horizon change the objective?

15 Limits of multiplicative functions on $\mathbb{N}$

A uniformly chosen large integer has nearly independent divisibility patterns at any fixed collection of primes. This suggests replacing its prime exponents by independent geometric variables. The essential second step is to control the contribution of the primes or divisors that were left out. The normalized divisor sum makes both steps explicit.

Write

$$\sigma(n) = \frac{1}{n}\sum_{d|n} d = \sum_{d|n}\frac{1}{d} = \prod_{p}\frac{1-p^{-v_p(n)-1}}{1-p^{-1}}. \tag{15.1}$$

Here σ denotes the normalized divisor sum; the more usual notation for it is $\sigma_1(n)/n$. If X_N is uniform on $\{1,\dots,N\}$, what is the limiting distribution of $\sigma(X_N)$?

For each fixed finite set of primes, their valuations in X_N converge jointly to independent variables Y_p with

$$\mathbb{P}(Y_p = k) = (1-p^{-1})p^{-k}, \qquad k = 0, 1, \dots. \tag{15.2}$$

This follows by counting residue classes modulo products of prime powers. The collection $(Y_p)_p$ describes a random divisibility pattern. It does not usually represent a finite integer: independence and $\sum_p 1/p = \infty$ imply that infinitely many Y_p are positive almost surely.

For a multiplicative function the factorization is

$$f(n) = \prod_p f(p^{v_p(n)}). \tag{15.3}$$

The more restrictive expression $\prod_p f(p)^{v_p(n)}$ requires complete multiplicativity. The natural proposed limit is $\prod_p f(p^{Y_p})$, when this product exists, together with a tail estimate allowing passage from finitely many primes to all primes.

For context, the Erdős–Wintner theorem concerns a real additive function g, meaning $g(mn) = g(m) + g(n)$ for coprime m, n. It says that $g(X_N)$ has a finite real limiting law exactly when the three series

$$\sum_{|g(p)|>1}\frac{1}{p}, \qquad \sum_{|g(p)|\le 1}\frac{g(p)^2}{p}, \qquad \sum_{|g(p)|\le 1}\frac{g(p)}{p} \tag{15.4}$$

converge. The last series need not converge absolutely. The limiting characteristic function is

$$\phi(t) = \prod_p (1-p^{-1})\sum_{\nu\ge 0}\frac{e^{itg(p^\nu)}}{p^\nu}. \tag{15.5}$$

These statements and effective forms are recorded by Tenenbaum and Verwee [13]. For positive multiplicative f, applying this to $g = \log f$ gives convergence to a positive random product under these conditions. A limit concentrated at zero for $f(X_N)$ is outside that finite-logarithm formulation.

For the normalized divisor sum, there is an elementary proof of convergence which also controls the neglected terms. Define

$$\Sigma = \prod_p \frac{1-p^{-Y_p-1}}{1-p^{-1}} = \sum_{d\ge 1}\frac{1\{v_p(d)\le Y_p \text{ for every } p\}}{d}. \tag{15.6}$$

Monotone convergence and independence give

$$\mathbb{E}\Sigma = \sum_{d\ge 1}\frac{1}{d^2} = \zeta(2) < \infty, \tag{15.7}$$

so the product is finite almost surely. Truncate both divisor sums at $d \le D$. The finite vectors of divisibility indicators converge by residue-class counting, while uniformly in N,

$$\mathbb{E} \sum_{\substack{d \mid X_N \\ d > D}} \frac{1}{d} = \sum_{D < d \le N} \frac{\lfloor N/d \rfloor}{Nd} \le \sum_{d > D} \frac{1}{d^2}. \tag{15.8}$$

The same tail bound holds for Σ. Truncation and Markov's inequality therefore prove $\sigma(X_N) \to_d \Sigma$ without invoking the full theorem.

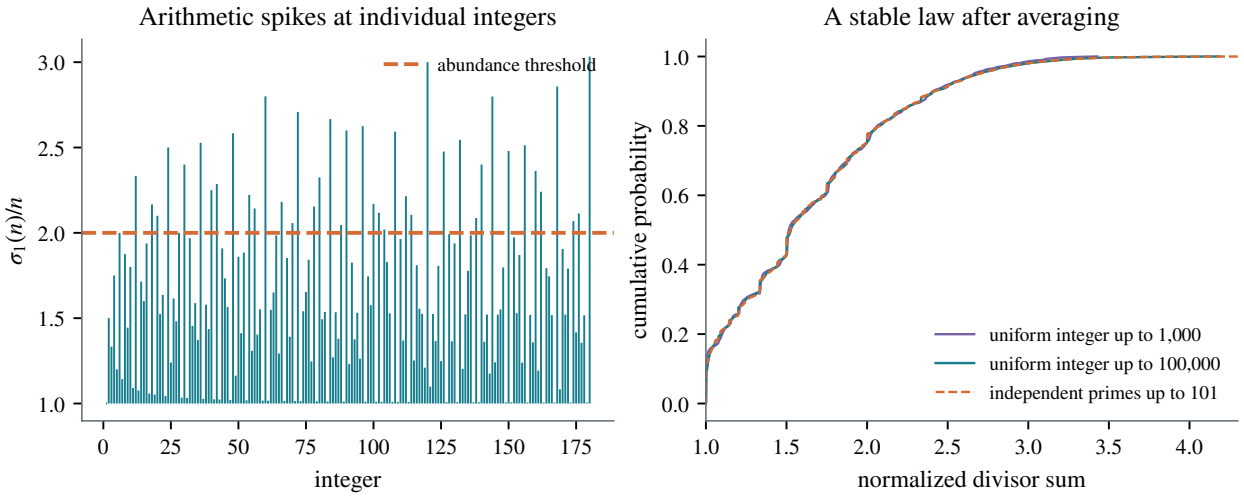


Figure 17: Left: the normalized divisor sum on the first 180 integers; values above two correspond to abundant integers. Right: the empirical laws for uniform integers up to 1000 and 100000, compared with 100000 samples of the independent prime product truncated at 101. The mean omitted logarithmic tail is at most $1/101$.

The second moment has a short divisor-pair calculation. The probability that the limiting divisibility pattern contains both a and b is $1/\operatorname{lcm}(a,b)$. Tonelli's theorem therefore gives

$$\mathbb{E}\Sigma^2 = \sum_{a,b \ge 1} \frac{1}{ab \operatorname{lcm}(a,b)} = \sum_{a,b \ge 1} \frac{\gcd(a,b)}{a^2 b^2}.$$

Write $a = du$, $b = dv$ with $\gcd(u,v) = 1$. Summing first over d yields $\zeta(3)$, while the coprime sum has Euler product $\prod_p (1 + p^{-2})/(1 - p^{-2}) = \zeta(2)^2/\zeta(4)$. Hence

$$\mathbb{E}\Sigma^2 = \frac{\zeta(2)^2 \zeta(3)}{\zeta(4)} \approx 3.0051423. \tag{15.9}$$

For higher moments, the prime factors are convenient. Since $\mathbb{E}p^{-jY_p} = (1 - p^{-1})/(1 - p^{-j-1})$, for any positive integer m the local mth moment is

$$p(1 - p^{-1})^{1-m} \sum_{j=0}^{m} (-1)^j \binom{m}{j} \frac{1}{p^{j+1} - 1}. \tag{15.10}$$

The product of these factors equals $\mathbb{E}\Sigma^m$ and is finite: for fixed m, each factor is $1 + O_m(p^{-2})$. Indeed, the factor differs from one only when $Y_p \ge 1$, an event of probability $1/p$, and then its value differs from one by $O_m(1/p)$.

The logarithm is an independent sum:

$$\log \Sigma = \sum_p \sum_{j\geq 1} \frac{p^{-j}}{j}(1 - p^{-jY_p}), \tag{15.11}$$

$$\mathbb{E} \log \Sigma = \sum_p \sum_{j\geq 1} \frac{p^j - 1}{j(p^{2j+1} - p^j)}. \tag{15.12}$$

For a single prime, the inner sum is $p^{-2} - \frac{1}{2}p^{-3} + O(p^{-4})$, giving a useful estimate for the tail.

This representation also gives a direct way to bracket abundance probabilities. For an integer cutoff $P \geq 2$, let Σ_P be the product over primes at most P and let $R_P = \log(\Sigma/\Sigma_P) \geq 0$. Since $\log[(1 - p^{-Y_p-1})/(1 - p^{-1})] \leq 1\{Y_p \geq 1\}/(p - 1)$,

$$\mathbb{E}R_P \leq \sum_{p>P} \frac{1}{p(p-1)} \leq \frac{1}{P}. \tag{15.13}$$

Thus for $u > 0$ and $\delta > 0$,

$$\mathbb{P}(\Sigma_P > u) \leq \mathbb{P}(\Sigma > u) \leq \mathbb{P}(\Sigma_P > ue^{-\delta}) + \frac{1}{P\delta}. \tag{15.14}$$

Finite-prime calculations can further truncate exponents, charging the omitted mass $\mathbb{P}(Y_p > K) = p^{-K-1}$ explicitly. Taking $u = 2$ gives bounds for the limiting probability of abundance. Sharper truncation and arithmetic estimates would be needed for high precision.

The moments do determine this law. A coarse estimate already proves it.

Proposition 15.1. *For every integer $m \geq 1$,*

$$\mathbb{E}\Sigma^m \leq Cm^m, \qquad C = \exp(e^2 - 1).$$

Consequently $\mathbb{E}e^{t\Sigma} < \infty$ for $0 \leq t < e^{-1}$, and the sequence of moments uniquely determines the distribution of Σ.

Proof. Write $F_p = (1 - p^{-Y_p-1})/(1 - p^{-1})$. The factors with $p \leq m$ satisfy

$$\prod_{p\leq m} F_p \leq \prod_{j=2}^{m} (1 - 1/j)^{-1} = m.$$

For $p > m$, the event $Y_p \geq 1$ has probability $1/p$. Also $(1 - 1/p)^{-m} \leq e^{2m/p}$, so convexity of e^x on $[0, 2]$ gives

$$\mathbb{E}F_p^m \leq 1 + \frac{e^{2m/p} - 1}{p} \leq 1 + (e^2 - 1)\frac{m}{p^2}.$$

Independence, $\sum_{j>m} j^{-2} \leq 1/m$, and monotone convergence give the moment bound. Since $m! \geq (m/e)^m$, the exponential series converges in expectation for $t < e^{-1}$. Any other probability law on $[0, \infty)$ with these moments has the same convergent moment generating function near zero and hence the same distribution. □

This answers the determinacy issue while leaving the numerical conditioning of moment-based inversion to be assessed separately.

Question 15.2. Which representations of this law give the most effective numerical bounds on $\mathbb{P}(\Sigma > u)$? How do finite-prime distribution calculations compare with moment methods and Fourier inversion of (15.5)?

16 Egyptian fractions: random inputs and the case $4/n$

Subtract the largest unit fraction that fits, then repeat. This greedy rule represents every rational number in $(0, 1)$ as a finite sum of distinct unit fractions. Its length is a random variable as soon as we randomize the input. There are two particularly natural experiments: choose a numerator uniformly over a fixed denominator, or keep the numerator equal to four and choose a large denominator. They lead to very different answers. In the first experiment even the shortest length diverges; in the second, almost every input needs only one or two terms, while the greedy algorithm has a positive chance of using four.

For $0 < a < b$, let $G(a/b)$ be the greedy length, and let $L(a/b)$ be the minimum number of terms in a representation

$$\frac{a}{b} = \frac{1}{d_1} + \cdots + \frac{1}{d_k}, \qquad 2 \le d_1 < \cdots < d_k.$$

These depend on the rational number, whether or not a/b is reduced. Write K_N for a uniform element of $\{1, \dots, N-1\}$ and set $G_N = G(K_N/N)$, $L_N = L(K_N/N)$.

16.1 Termination and a short constructive alternative

The greedy step chooses $d = \lceil b/a \rceil$ and leaves

$$\frac{a}{b} - \frac{1}{d} = \frac{ad - b}{bd}, \qquad 0 \le ad - b < a. \tag{16.1}$$

Even without cancellation, the positive integer numerator strictly decreases. Thus $G(a/b) \le a$. The new remainder is less than $1/[d(d-1)]$, so subsequent denominators strictly increase. This is the classical Fibonacci–Sylvester algorithm; Eppstein's account [14] compares it with several alternatives.

The numerator bound is generous. A different construction already gives a logarithmic bound for the optimum.

Proposition 16.1. *For $0 < a < b$, $L(a/b) \le 2\lceil \log_2 b \rceil$.*

Proof. Put $m = \lceil \log_2 b \rceil$, $Q = 2^m$, and divide $Qa = bs + r$ with $0 \le r < b$. Then

$$\frac{a}{b} = \frac{s}{Q} + \frac{r}{bQ}.$$

Both s and r are smaller than Q. Their binary expansions give at most m reciprocals each, with denominators $Q/2^j$ and $bQ/2^j$, respectively. Each list has distinct entries. An intersection between the lists would force b to be a power of two, in which case $r = 0$. □

This binary construction illustrates the classical divisor method: write a numerator as a short sum of divisors of a convenient auxiliary denominator. It proves sublinear optimal length without using a deep existence theorem.

The literature gives substantially shorter expansions. Vose [15] proved a uniform $O(\sqrt{\log b})$ bound. A 2026 result reaches the double-logarithmic order proposed by Erdős:

Theorem 16.2 (Uniform short expansions, 2026)**.** *There is an absolute constant C such that every $0 < a < b$ satisfies*

$$L(a/b) \le C \log\log(3b).$$

This is Theorem 1.1 of [16], with a human-written exposition and simplification in Bloom–van Doorn [17, Theorem 1]. The latter proof permits fractions whose numerators divide an auxiliary integer, extending the divisor method. Smooth-number residue sets, a finite-field sum-product theorem, and representations by primes supply these fractions. A greedy prefix reduces the general input to a small-numerator range. The theorem is an external input here; the probabilistic consequences below have short proofs of their own.

16.2 A typical numerator really needs a growing number of terms

A uniform upper bound alone could be driven by rare numerators. To see why the double logarithm appears for most inputs, consider the geometry of numbers representable by a bounded number of unit fractions.

Proposition 16.3. *For every integer $m \geq 0$ and $N \geq 2$,*

$$\mathbb{P}(L_N \leq m) \leq \frac{N}{N-1}(m+2)^2(2N)^{-2^{-m}}. \tag{16.2}$$

In particular, for each $0 < \eta < 1$,

$$\mathbb{P}\big(L_N \leq (1-\eta)\log_2\log N\big) \longrightarrow 0, \qquad \liminf_{N\to\infty} \frac{\mathbb{E}L_N}{\log_2\log N} \geq 1.$$

Proof. Let S_m be the set of sums of at most m reciprocals of positive integers, allowing repetitions, and put $S_0 = \{0\}$. These are compact sets, since $\{0, 1, 1/2, 1/3, \ldots\}$ is compact and S_m is its m-fold sum. Write $v_m(\varepsilon)$ for the Lebesgue measure of the ε-neighborhood of S_m in $\mathbb{R}$.

For an integer $A \geq 1$, either every denominator exceeds A, giving a sum at most $m/(A+1)$, or some term is $1/d$ with $d \leq A$. Hence

$$S_m \subset [0, m/(A+1)] \ \cup \bigcup_{d=1}^{A} (1/d + S_{m-1}).$$

It follows that

$$v_m(\varepsilon) \leq m/(A+1) + 2\varepsilon + A v_{m-1}(\varepsilon).$$

For $0 < \varepsilon \leq 1$, induction with $A = \lfloor \varepsilon^{-2^{-m}} \rfloor$ gives

$$v_m(\varepsilon) \leq (m+2)^2 \varepsilon^{2^{-m}}.$$

Indeed $v_0 = 2\varepsilon \leq 4\varepsilon$, and the next induction step has coefficient at most $m+2+(m+1)^2 \leq (m+2)^2$.

The intervals of radius $1/(2N)$ around the grid points k/N are disjoint up to endpoints. Those with $L(k/N) \leq m$ lie in the $1/(2N)$-neighborhood of S_m. Their total length proves (16.2). For $m = \lfloor (1-\eta)\log_2\log N \rfloor$, the negative logarithm of $(2N)^{-2^{-m}}$ grows at least as fast as $(\log N)^{\eta}$; the polynomial factor in m is negligible. Finally use $\mathbb{E}L_N \geq m\mathbb{P}(L_N > m)$ and let η decrease to zero. □

Combining this elementary estimate with Theorem 16.2 yields

$$\mathbb{E}L_N = \Theta(\log\log N). \tag{16.3}$$

Both L_N and G_N tend to infinity in probability. Thus neither has a limiting probability law on the positive integers without a change of scale, and both expectations diverge. The interesting limiting-law question concerns their centered or rescaled versions.

16.3 The mean greedy length is sublinear

The elementary bound $G_N \le K_N$ gives $\mathbb{E}G_N \le N/2$. A continuous-input calculation improves this to a strict order statement.

Proposition 16.4. *For the uniform numerator experiment, $\mathbb{E}G_N = o(N)$.*

Proof. Run the greedy algorithm on a real $x \in (0,1)$, writing $r_j(x)$ for the remainder after j steps and $D_j(x)$ for the product of the first j denominators. Put $A_j = D_j r_j$, with $A_0 = x$, and set $A_j = 0$ after termination. Equation (16.1) gives $0 \le A_j \le x$.

Condition a uniform real input on a particular finite greedy prefix. Its possible values form an interval $(s, s+\ell)$, up to endpoints, where s is the sum of that prefix. Thus its residual is uniform on $(0,\ell)$ and D_j is constant. On a next-denominator branch $d \ge 2$, the residual ranges from $1/d$ to some $b \le 1/(d-1)$. The ratio of the integrals of the new and old virtual numerators is

$$\frac{\int_{1/d}^{b}(dr-1)\,dr}{\int_{1/d}^{b} r\,dr} = d - \frac{2}{b+1/d} \le \frac{d}{2d-1} \le \frac{2}{3}.$$

Sum over the branches and then over the prefixes. For uniform $U \in (0,1)$ this proves

$$\mathbb{E}A_j(U) \le \tfrac{1}{2}(2/3)^j.$$

For a grid input $x = k/N$, the integer $NA_j(x)$ is precisely the unreduced numerator after j steps. The remaining number of steps is at most that numerator, so $G(k/N) \le j + NA_j(k/N)$. For fixed j, A_j is bounded and continuous at every irrational input: a finite greedy prefix is locally constant there. Its discontinuities therefore form a set of measure zero, and its grid averages converge to its integral. Consequently

$$\limsup_{N\to\infty} \frac{\mathbb{E}G_N}{N} \le \int_0^1 A_j(x)\,dx \le \tfrac{1}{2}(2/3)^j.$$

Let $j \to \infty$. □

The order question is now between a double-logarithmic lower bound and a sublinear upper bound. Eppstein [14] describes a random-remainder heuristic suggesting logarithmic greedy length; the remainders are arithmetic quantities, so that heuristic needs an actual distributional argument. Figure 18 also shows why the denominator's factorization should be tracked in experiments.

Question 16.5. Is $\mathbb{E}G_N = O(\log N)$ uniformly over denominators? Does $L_N/\log\log N$ converge in probability to a constant, and can the constant depend on the arithmetic sequence of denominators? What centering and scaling give a nondegenerate law for G_N? These finer questions are left unresolved here; no claim about their priority is made.

16.4 Fixed numerator: a periodic greedy law

Keeping the numerator fixed changes the problem. Its greedy length is bounded, and the entire distribution is computable by a finite residue calculation.

Proposition 16.6. *For fixed $a \ge 1$, $g_a(n) := G(a/n)$, $n > a$, is periodic in n with period $a!$. Hence the law for a uniform $n \in \{a+1,\dots,N\}$ has a computable limit with rational probabilities.*

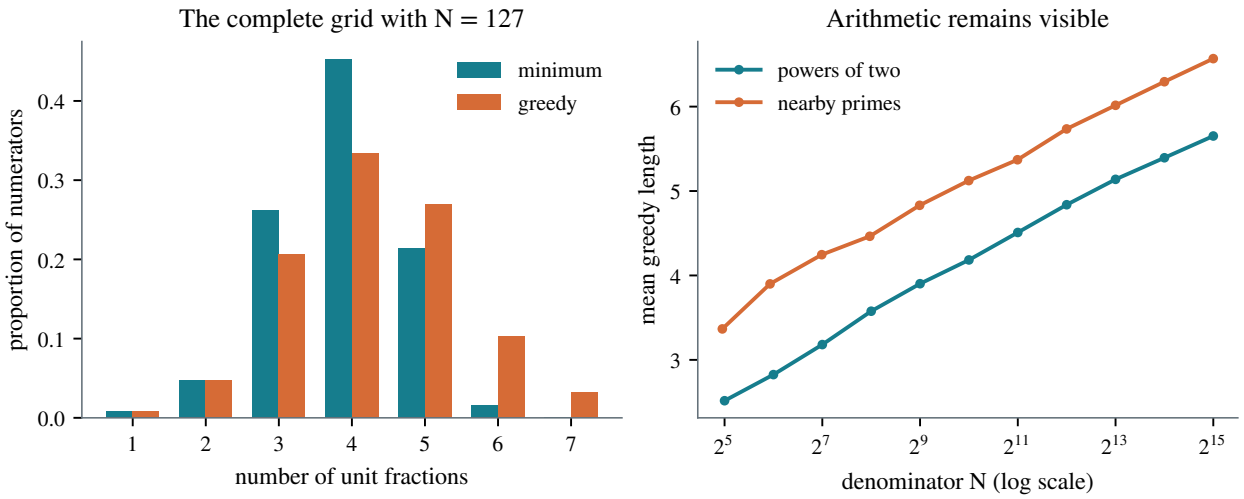


Figure 18: Complete numerator censuses, using exact integer arithmetic. Left: greedy and minimum lengths for $k/127$, $1 \le k < 127$; minimum lengths come from exhaustive finite searches. Right: mean greedy lengths for powers of two and the largest primes below them. These are finite calculations, with no simulation error or censored inputs.

Proof. Induct on a. The claim for $a = 1$ is immediate. If $a \mid n$, the length is one. Otherwise put $r = n \bmod a$, $c = a - r < a$. The first remainder is

$$\frac{c}{b(n)}, \qquad b(n) = \frac{n(n+c)}{a}, \quad g_a(n) = 1 + g_c(b(n)).$$

On replacing n by $n + a!$, the remainder class r stays fixed, and

$$b(n + a!) - b(n) = (a-1)!\,(2n + c + a!),$$

which is divisible by $c!$. The induction hypothesis proves periodicity. Uniform residue frequencies differ from $1/a!$ by $O(1/N)$, completing the limiting-law assertion. □

The factorial is a convenient period, not a claim about the least period. One can compute the law by applying the integer recursion to any $a!$ consecutive denominators larger than a.

16.5 Four over a random denominator

For $a = 4$ the residue calculation is small enough to display completely.

Theorem 16.7. *For every $n > 4$,*

$$G(4/n) = \begin{cases} 1, & n \equiv 0 \pmod 4, \\ 3, & n \equiv 5, 13 \pmod{24}, \\ 4, & n \equiv 1, 17 \pmod{24}, \\ 2, & \textit{otherwise.} \end{cases}$$

For a uniform $n \in \{5, \dots, N\}$, its limiting distribution on $\{1, 2, 3, 4\}$ is $(1/4, 7/12, 1/12, 1/12)$, and its mean tends to 2.

Proof. Multiples of four give a unit fraction. For $n \equiv 2 \pmod 4$, reduction leaves $2/q$ with q odd, requiring two terms. For $n \equiv 3 \pmod 4$, the first remainder already has numerator one.

For $n \equiv 1 \pmod 4$, the first remainder is $3/B$, where $B = n(n+3)/4$. If $3 \mid n$, then $3 \mid B$, so the total is two terms. Otherwise $B \equiv 1 \pmod 3$, and the next remainder is $2/[B(B+2)/3]$. This is a unit fraction when B is even, and needs two terms when B is odd. The first case is $n \equiv 5 \pmod 8$, the second $n \equiv 1 \pmod 8$. Combining the moduli gives the displayed classes. Their counts in a full period of 24 are $6, 14, 2, 2$. □

The shortest length has a simpler limiting law, for an arithmetic reason. The following two-term criterion is classical; see the discussion in Bloom–Elsholtz [18].

Proposition 16.8. *For $n > 4$, $L(4/n) = 1$ exactly when $4 \mid n$. Moreover,*

$$L(4/n) \geq 3 \quad \Longleftrightarrow \quad n \text{ is odd and every prime divisor of } n \text{ is } 1 \pmod 4.$$

For a uniform $n \in \{5, \dots, N\}$, the limiting law on $\{1, 2, 3, 4\}$ is $(1/4, 3/4, 0, 0)$, with limiting mean $7/4$.

Proof. The one-term assertion follows from $4/n = 1/d$. The case $n \equiv 2 \pmod 4$ was already settled by the greedy calculation. If an odd n has a prime divisor $p \equiv 3 \pmod 4$, the identity

$$\frac{4}{p} = \frac{1}{(p+1)/4} + \frac{1}{p(p+1)/4}$$

scales to a two-term representation of $4/n$ with distinct denominators. Conversely, for odd n a two-term representation would give

$$(4x - n)(4y - n) = n^2.$$

Both factors are positive. If all prime divisors of n are $1 \pmod 4$, then $n \equiv 1 \pmod 4$, so both factors are $3 \pmod 4$, whereas every divisor of n^2 is $1 \pmod 4$. This is impossible.

The exceptional integers have density zero. For any finite set of primes $p \equiv 3 \pmod 4$, their upper density is at most $\prod_p (1 - 1/p)$. The divergence of the reciprocal-prime sum in this progression makes these products tend to zero. Thus only the one- and two-term cases survive in the limiting distribution. Since $L(4/n) \leq 4$, convergence of the mean follows as well. □

These limits hold without assuming the Erdős–Straus conjecture. That conjecture asks for three terms for *every* denominator, whereas the uniform limiting law loses the exceptional set altogether. More generally, for each fixed $a \geq 2$, the limiting shortest-length law is $(1/a, 1 - 1/a)$ on $\{1, 2\}$. Indeed, almost every integer has a prime divisor $p \equiv -1 \pmod a$, $p > a$, and

$$\frac{a}{p} = \frac{1}{(p+1)/a} + \frac{1}{p(p+1)/a}.$$

Scale this identity and use the reciprocal-prime divergence in that progression. One-term representations occur exactly when $a \mid n$.

The rare cases retain substantial structure. Vaughan [19] proved that the number of $n \leq N$ for which $4/n$ has no three-term representation is at most

$$N \exp\{-c(\log N)^{2/3}\} \tag{16.4}$$

for some $c > 0$ and sufficiently large N. Pomerance–Weingartner [20, Theorem 1.3] obtain an extension uniform over a range of numerators. For fractions less than one, allowing repeated

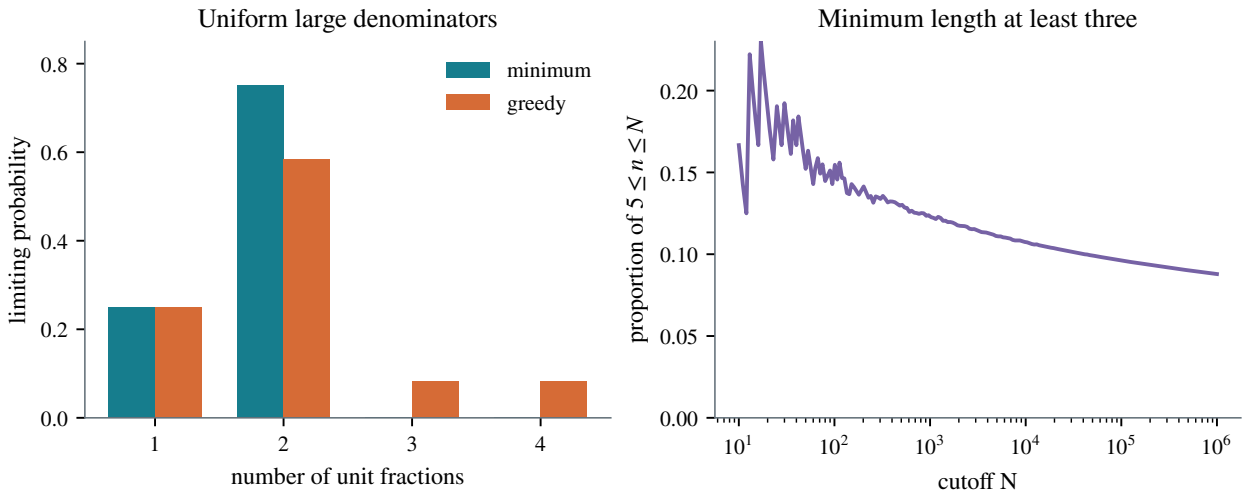


Figure 19: The fixed numerator 4 has explicit limiting laws. Left: the proved limiting probabilities for greedy and minimum length. Right: the exact finite proportion needing at least three terms, obtained by sieving out even integers and multiples of primes 3 (mod 4). The vanishing exceptional proportion can remain visible in finite samples.

denominators does not change the existence of a representation with at most a specified number of terms: Takenouchi's conversion theorem supplies distinct ones [17, Appendix A].

One useful conditioning keeps this exceptional arithmetic visible. Choose p uniformly among primes at most N with $p \equiv 1 \pmod 4$. Then $L(4/p) \ge 3$, and (16.4), divided by the number of these primes, proves $\mathbb{P}(L(4/p) = 3) \to 1$. By contrast, the prime number theorem in progressions modulo 24 and Theorem 16.7 give

$$\mathbb{P}(G(4/p) = 3) \longrightarrow \tfrac{1}{2}, \qquad \mathbb{P}(G(4/p) = 4) \longrightarrow \tfrac{1}{2}.$$

Thus the greedy algorithm wastes a term for asymptotically half of these prime inputs, even though three terms suffice for almost all of them.

Question 16.9. Among integers whose prime factors are all 1 (mod 4), what is the limiting proportion with greedy length four? Can explicit three-term identities describe most of those inputs with controlled denominators? This conditional distribution keeps the part of the problem that the uniform integer limit discards.

16.6 What the finite computations certify

Minimum length is computable by a complete finite search. If a positive remainder r is to be written with at most k increasing denominators, its first denominator lies between $\lceil 1/r \rceil$ and $\lfloor k/r \rfloor$. Recurse over that finite interval, requiring each new denominator to exceed the preceding one. For two terms, factoring

$$(ax - b)(ay - b) = b^2$$

tests $a/b = 1/x + 1/y$ exactly. Searching lengths in increasing order up to a known construction proves minimality, without an arbitrary cutoff on the largest denominator. The search is finite but can be expensive.

The accompanying code records full numerator censuses and exact representations. For example, at $N = 127$ the mean minimum length is $487/126 \approx 3.8651$, while at $N = 128$ it is $386/127 \approx 3.0394$.

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